



Modelling Financial Markets

Seminar 1: Stylised Facts and Returns — Hands-on



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Acronyms used in this material (1/2)

- ▣ **ACF**: AutoCorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARCH-LM**: ARCH Lagrange Multiplier (test)
- ▣ **BET-TR**: BET Total Return (index)
- ▣ **BNR**: Banca Națională a României — *the National Bank of Romania*
- ▣ **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- ▣ **CAGR**: Compound Annual Growth Rate
- ▣ **CI**: Confidence Interval
- ▣ **COVID-19**: COronaVirus Disease 2019
- ▣ **CUSUM**: CUmulative SUM (of squares)
- ▣ **EGARCH**: Exponential GARCH
- ▣ **ETF**: Exchange-Traded Fund
- ▣ **FIGARCH**: Fractionally Integrated GARCH
- ▣ **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)

Acronyms used in this material (2/2)

- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HAR**: Heterogeneous AutoRegressive (model)
- ▣ **JB**: Jarque–Bera (test)
- ▣ **MDD**: Maximum DrawDown
- ▣ **OHLC**: Open, High, Low, Close
- ▣ **OLS**: Ordinary Least Squares
- ▣ **QQ**: Quantile–Quantile (plot)
- ▣ **RMSE**: Root Mean Squared Error
- ▣ **SE**: Standard Error
- ▣ **US**: United States
- ▣ **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ **VIX**: CBOE Volatility Index

Today's Question and Seminar Structure (1/3)

- Question of the seminar
 - ▶ **which stylised facts of returns survive inference that allows for heavy tails and volatility clustering?**
- Three types of exercises
 - ▶ **A:** calculation exercises, on paper
 - ▶ **B:** exercises on real data, solved in the notebook
 - ▶ **C:** open questions, e.g. correcting an answer given by an AI tool
- Each exercise is marked
 - ▶ **[Solved]:** the full solution is presented in the seminar
 - ▶ **[Proposed]:** you solve it; the solution is discussed in the seminar

Today's Question and Seminar Structure (2/3)

- Today's exercises, in order
 - ▶ A1: aggregating returns over time and annualisation
 - ▶ A2(a): the depth of a drawdown
 - ▶ B1: descriptive statistics and tail thickness for five markets
 - ▶ A3: the Jarque–Bera and Ljung–Box tests, step by step
 - ▶ B2: heavy tails and VaR 1% for BET-TR
 - ▶ A4: the kurtosis of a mixture of Normal distributions
 - ▶ B3: uncorrelated but not independent returns (Bitcoin)
 - ▶ B4: the Ljung–Box test robust to volatility clustering
 - ▶ B5: aggregating returns and approaching the Normal distribution
 - ▶ C1: correcting an AI answer with errors
- After lecture 1 (they use the Hill estimator, presented in the lecture)
 - ▶ A5 and B6: the Hill estimator of the tail index
 - ▶ A6: the widening of the autocorrelation band under volatility clustering

Today's Question and Seminar Structure (3/3)

- Outputs to produce
 - ▶ the answers to the A exercises
 - ▶ one figure and one table per B exercise
 - ▶ the corrected C1 code
 - ▶ one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#)
 - ▶ each task names its notebook section
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro; Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A1	how do returns add up over time?	Solved	–
A2(a)	how deep is a drawdown?	Proposed	A1
B1	how heavy are the tails, and how fragile is the kurtosis?	Solved	–
A3	from a statistic to a p-value: JB and Ljung–Box	Solved	–
B2	how large is the VaR 1% (Value-at-Risk) of BET-TR?	Proposed	B1
A4	why does random volatility create heavy tails?	Solved	–
B3	are Bitcoin returns independent?	Solved	–
B4	which autocorrelations survive robust inference?	Solved	–
B5	do returns approach the Normal distribution with the horizon?	Proposed	B1, B3
C1	what is wrong in an AI answer?	Proposed	A3, B3, B4
After the lecture	A5 [Solved], B6 [Solved], A6 [Proposed] (tail heaviness, width of the ACF band)	homework	A4, B4
Extension packs	A1(c), A2(b–e), A7–A11, B1(d), B7–B15, C2, C3 (optional)	Solved or Proposed (marked on each)	named on each task

- ▣ **[Solved]**: full solution in the slides and in the notebook, a model to copy
- ▣ **[Proposed]**: you solve it, the solution is discussed in class
- ▣ Part A: derivations on paper
- ▣ Part B: verification and inference on real data, each ending with an interpretation question
- ▣ Part C: open questions and AI critique

Data Used

Series	Price used	Period	Calendar	Returns
BET-TR	daily close	23 Sep 2014 – 18 Sep 2026	BVB trading days	2 992
Bitcoin	daily close (USD)	17 Sep 2014 – 18 Sep 2026	every day, 24:00 UTC	4 384
EUR/RON	BNR reference rate	4 Jul 2005 – 18 Sep 2026	Romanian working days	5 352
S&P 500	daily OHLC	2 Jan 1990 – 18 Sep 2026	US trading days	9 245
SPY ETF	OHLC, adjusted close (for dividends and splits)	2 Jan 2008 – 18 Sep 2026 (B15)	US trading days	–
VIX, 9-day VIX	daily close	C3 only	US trading days	–

- Daily prices of the course data, one series per asset (field used: close / adjusted close)
- EUR/RON is the official BNR reference rate, not a series of market quotes
- Units: returns in decimals in the code, reported in % on the slides
- Calendars: each series on its own calendar
 - ▶ multi-series analyses first align the *prices* on common days

Setup: Run and Verify

```
# notebook section "Data" defines load_data(name)
closes = {a: load_data(a)['Close'] for a in
           ['sp500', 'bettr', 'btc', 'eurron']}
rets = {a: np.log(c).diff().dropna()
        for a, c in closes.items()}
r_bet, r_btc = rets['bettr'], rets['btc']
r_eur, r_sp = rets['eurron'], rets['sp500']
for a, r in rets.items():
    print(a, r.index[0].date(), len(r))
```

- Expected output: first return date and count

Series	first return	T
S&P 500	1990-01-03	9 245
BET-TR	2014-09-24	2 992
Bitcoin	2014-09-18	4 384
EUR/RON	2005-07-05	5 352

- Conventions in the code
 - moments: `scipy.stats.skew` and `kurtosis` (excess kurtosis, no small-sample correction)
 - standard deviations with `ddof=1`
 - quantiles with `numpy.quantile` (linear)
- If a count differs, stop and check the series, the price type and the start date

Notation Used in Every Exercise (1/7): Prices and Returns

- Indices and prices

- ▶ $t = 0, 1, \dots, T$: the trading-day index; $t - 1$ is the previous day
- ▶ P_t : the closing price on day t ; T : the number of returns

- Simple return**: the relative price change over one day

$$R_t = \frac{P_t}{P_{t-1}} - 1$$

- Log return**: the natural logarithm ($\ln$) of the price ratio

$$r_t = \ln \frac{P_t}{P_{t-1}} = \ln(1 + R_t)$$

- ▶ $r_t \approx R_t$ for small moves; $r_t < R_t$ whenever $R_t \neq 0$
- ▶ one worked line: $P_0 = 100$, $P_1 = 102$: $R_1 = 2.00\%$, $r_1 = \ln 1.02 = 1.98\%$

Notation Used in Every Exercise (2/7): Mean, Standard Deviation, Standardisation

- Writing conventions

- ▶ the hat: $\hat{\theta}$ is the sample estimate of a population quantity θ
- ▶ $\sum_t$: the sum over all days $t = 1, \dots, T$

- **Sample mean** $\bar{r} = \frac{1}{T} \sum_t r_t$: the average daily return

- **Sample standard deviation** (daily volatility): the typical spread of returns around the mean

$$\hat{\sigma} = \sqrt{\frac{1}{T-1} \sum_t (r_t - \bar{r})^2}$$

- Demeaned return $x_t = r_t - \bar{r}$: the deviation from the mean

- Standardised return $z_t = (r_t - \bar{r})/\hat{\sigma}$: the number of standard deviations from the mean

- ▶ $z_t = -3$: a day three standard deviations below the mean

- **i.i.d.**: independent and identically distributed, the reference assumption of the classical tests

- ▶ today's return does not depend on past returns, and its distribution is the same every day

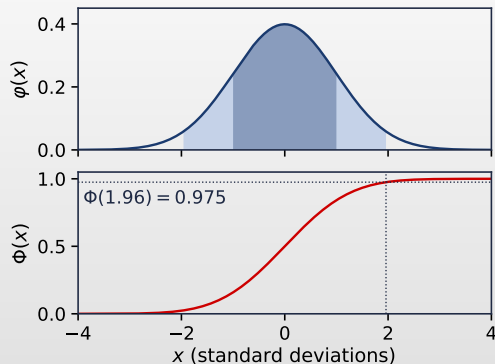
Notation Used in Every Exercise (3/7): Probability Operators and the Normal Distribution

- ▣ Probability operators, for a random variable X
 - ▶ $\mathbb{E}X$: the population mean (expectation); $\text{Var}(X) = \mathbb{E}(X - \mathbb{E}X)^2$: the variance
 - ▶ $\text{Cov}(X, Y)$, $\text{corr}(X, Y)$: the covariance and the correlation of two variables
 - ▶ $P(\cdot)$: the probability of an event
- ▣ **Normal distribution** $N(\mu, \sigma^2)$: mean μ , variance σ^2
 - ▶ $N(0, 1)$: the standard Normal distribution, with density and distribution function

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad \Phi(x) = P(Z \leq x) = \int_{-\infty}^x \varphi(u) du$$

Notation Used in Every Exercise (4/7): The Normal Curve

- $Z \sim N(0, 1)$
 - ▶ $\varphi(x)$: the height of the curve at x
 - ▶ $\Phi(x)$: the area under the curve to the left of x
- Shaded areas in the chart
 - ▶ $P(|Z| \leq 1) = 68\%$
 - ▶ $P(|Z| \leq 1.96) = 95\%$
- Φ^{-1} : the inverse of Φ , i.e. the $N(0, 1)$ quantiles
 - ▶ $\Phi^{-1}(0.975) = 1.96$, so Z lies in $[-1.96, 1.96]$ with probability 95%
- Standardisation: if $X \sim N(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma \sim N(0, 1)$



- | | | | |
|--|--|---|---|
| $P(Z \leq 1.96) = 95\%$ | $P(Z \leq 1) = 68\%$ | density $\varphi(x)$ | CDF $\Phi(x) = P(Z \leq x)$ |
|--|--|---|---|

Notation Used in Every Exercise (5/7): Aggregation over Time and Annualisation

- ▣ **Aggregation over time:** the return over days $1, \dots, n$, from P_0 to P_n

$$1 + R_{0 \rightarrow n} = \frac{P_n}{P_0} = \prod_{t=1}^n (1 + R_t), \quad r_{0 \rightarrow n} = \ln \frac{P_n}{P_0} = \sum_{t=1}^n r_t$$

- ▶ $\prod$: the product of the factors; simple returns compound (the factors $1 + R_t$ multiply)
- ▶ log returns add up: $\sum_{t=1}^n (\ln P_t - \ln P_{t-1}) = \ln P_n - \ln P_0$, because the intermediate terms cancel ($+\ln P_1$ with $-\ln P_1$, etc.)
- ▶ across assets, in a portfolio, it is the simple returns that add up with weights (A1(c))

- ▣ **Annualisation:** A is the number of return intervals in a year

$$\bar{r}_{\text{ann}} = A \bar{r}, \quad \hat{\sigma}_{\text{ann}} = \sqrt{A} \hat{\sigma}, \quad \text{because } \text{Var}\left(\sum_{t=1}^A r_t\right) = A\sigma^2$$

- ▶ $A = 252$ (BET-TR, S&P 500, EUR/RON: trading days), $A = 365$ (Bitcoin, traded every day)
- ▶ $A\sigma^2$ assumes uncorrelated returns with constant variance (A7: the autocorrelated case)
- ▶ example: $\hat{\sigma} = 1\%$ a day and $A = 252$: $\hat{\sigma}_{\text{ann}} = 15.87 \times 1\% = 15.9\%$

Notation Used in Every Exercise (6/7): Skewness, Kurtosis, Autocorrelation

- **Skewness** $\hat{S} = \frac{1}{T} \sum_t z_t^3$: the mean cube of the standardised returns
 - ▶ $\hat{S} = 0$ for a symmetric distribution
 - ▶ $\hat{S} < 0$: extreme losses are larger, in absolute value, than extreme gains
- **Kurtosis** $\hat{K} = \frac{1}{T} \sum_t z_t^4$: the mean fourth power, driven mostly by the extreme days
 - ▶ $\hat{K} = 3$ for the Normal distribution
 - ▶ **excess kurtosis** $\hat{K} - 3$: 0 for the Normal distribution, positive for heavy tails
- **Autocorrelation** at lag k : the correlation between returns k days apart

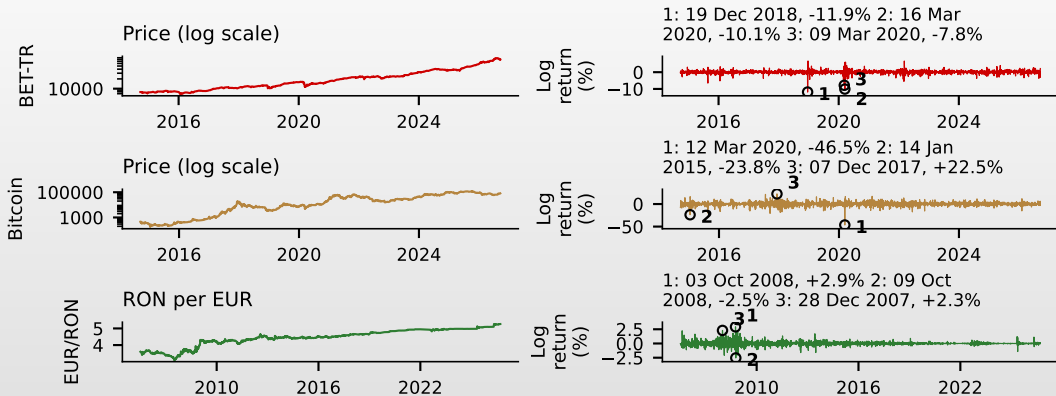
$$\hat{\rho}_k = \frac{\sum_{t=k+1}^T x_t x_{t-k}}{\sum_{t=1}^T x_t^2}$$

- ▶ $k = 1, 2, \dots$: the lag, in days; $\hat{\rho}_k \in [-1, 1]$; ρ_k without a hat: the population value
- ▶ $\hat{\rho}_k > 0$: a day above the mean tends to be followed, k days later, by a day above the mean
- ▶ $\hat{\rho}_k \approx 0$: no linear link between returns k days apart

Notation Used in Every Exercise (7/7): Statistical Tests

- The parts of a test
 - ▶ H_0 : the null hypothesis, for example “returns are not autocorrelated”
 - ▶ test statistic: a number computed from the data, larger the more the data contradict H_0
 - ▶ reference distribution: the distribution of the statistic when H_0 is true
- **The χ_m^2 distribution** (chi-squared with m degrees of freedom)
 - ▶ the distribution of the sum of squares of m independent $N(0, 1)$ variables; its mean is m
 - ▶ 5% critical values: 3.84 ($m = 1$), 5.99 ($m = 2$), 11.07 ($m = 5$), 18.3 ($m = 10$)
- **Critical value** at 5%: the value exceeded with probability 5% under H_0
 - ▶ reject H_0 at the 5% level if the statistic exceeds the critical value
- **p-value**: probability, under H_0 , of a statistic at least as extreme as the one observed
 - ▶ $p < 0.05$ is the same as exceeding the 5% critical value
 - ▶ it is not the probability that H_0 is true

First Look: Prices, Returns and Dated Extremes (1/2)



First Look: Prices, Returns and Dated Extremes (2/2)

- Every extreme day is a dated event
 - ▶ the bank-asset tax (19 Dec 2018)
 - ▶ the COVID-19 crash (March 2020)
 - ▶ the 2008 depreciation of the leu
- Calm and turbulent periods alternate in all three series
- EUR/RON is almost flat after 2013
- Look first, test later

 MFM_ch1_seminar_tests

Prerequisites for Today (1/23): The Stylised Facts

- ▣ **Stylised fact:** a statistical property shared by many assets, markets and periods [Cont \(2001\)](#)
- ▣ Today's four stylised facts
 - ▶ **heavy tails:** extreme days are more frequent than under the Normal distribution with the same variance
 - ▶ **volatility clustering:** large moves follow large moves, of either sign
 - ▶ **no linear autocorrelation:** today's return is not a linear function of past returns, $\hat{\rho}_k \approx 0$
 - ▶ **aggregational Gaussianity:** returns over longer horizons have tails closer to those of the Normal distribution

Prerequisites for Today (2/23): The Random Walk and Brownian Motion

- **Random walk:** the log price, $X_t = \ln P_t$, adds up independent shocks

$$X_t = X_{t-1} + r_t, \quad r_t \text{ i.i.d. } N(\mu\Delta, \sigma^2\Delta)$$

- ▶ Δ : the length of one step, in years (one trading day: $\Delta = 1/252$)
- ▶ μ : the annual drift of X_t ; σ : the annual volatility
- ▶ the benchmark against which the stylised facts are defined: i.i.d. returns with the Normal distribution

- **Standard Brownian motion** W_t , $t \geq 0$: the limit of the random walk as $\Delta \rightarrow 0$

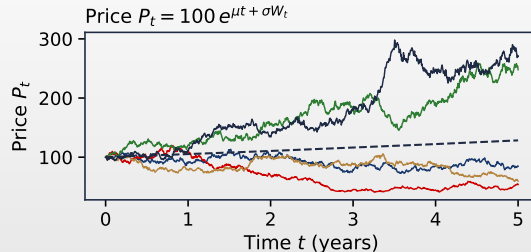
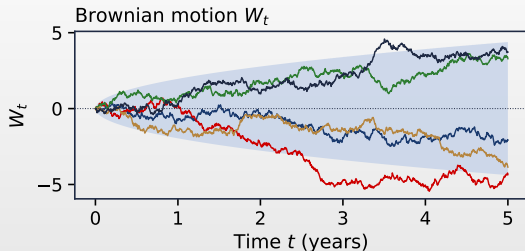
- ▶ $W_0 = 0$; the increment $W_t - W_s$, $s < t$, is independent of the past and has the distribution $N(0, t - s)$
- ▶ hence $W_t \sim N(0, t)$: the standard deviation grows like $\sqrt{t}$, and $|W_t| \leq 1.96\sqrt{t}$ with probability 95%

- **The price** as geometric Brownian motion ([A2](#), [A8](#))

$$X_t = X_0 + \mu t + \sigma W_t, \quad P_t = P_0 e^{\mu t + \sigma W_t}$$

- ▶ $P_t > 0$; the log return over an interval of length Δ has the distribution $N(\mu\Delta, \sigma^2\Delta)$

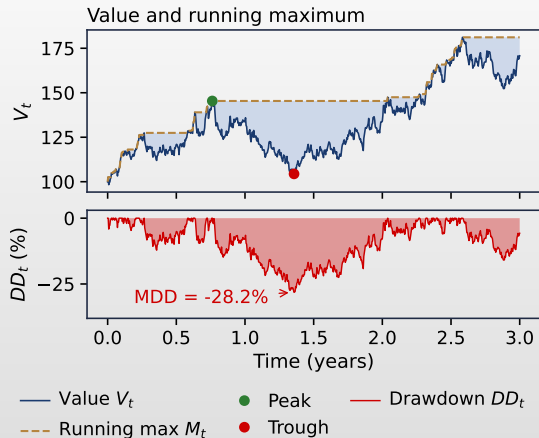
Prerequisites for Today (3/23): Simulated Brownian Paths



95% band $\pm 1.96\sqrt{t}$ — Simulated paths --- Median price $100e^{\mu t}$

- Left: five simulated paths of W_t over 5 years and the 95% band $\pm 1.96\sqrt{t}$, which widens like $\sqrt{t}$
- Right: the same paths turned into prices, $P_t = 100 e^{\mu t + \sigma W_t}$, with $\mu = 0.05$ and $\sigma = 0.20$
 - ▶ after 5 years the simulated prices lie between 54 and 270: uncertainty grows with the horizon
 - ▶ the dashed line $100 e^{\mu t}$ is the median price, because the median of W_t is 0

Prerequisites for Today (4/23): The Drawdown



- Running peak: the highest value reached up to time t

$$M_t = \max_{s \leq t} V_s$$

- V_t : the portfolio value or price at time t

- Drawdown: the relative loss from the running peak

$$DD_t = \frac{V_t}{M_t} - 1 \leq 0$$

- $DD_t = 0$ at every new high; the plot of DD_t is the underwater curve

- Maximum drawdown $MDD = \min_t DD_t$

- the largest fall from a peak to a later trough
- in the chart: -28.2%, from the green dot to the red dot

- CAGR = $(V_n/V_0)^{1/Y} - 1$: the constant annual growth rate

- Y : the number of years between V_0 and V_n

Prerequisites for Today (5/23): The Sharpe, Sortino and Calmar Ratios

- ▣ **Sharpe ratio**: the annual excess return per unit of annual volatility

$$SR = \frac{A \bar{R} - r_f}{\sqrt{A} \hat{\sigma}_R}$$

- ▶ $\bar{R}$, $\hat{\sigma}_R$: the mean and standard deviation of simple returns; r_f : the annual risk-free rate
- ▶ $SR = 0.5$: each percentage point of annual volatility earns 0.5 percentage points of excess return

- ▣ **Sortino ratio** Sortino & van der Meer (1991): penalises only the losing days

$$\text{Sortino} = \frac{A \bar{R} - r_f}{\sqrt{A} \hat{\sigma}_D}, \quad \hat{\sigma}_D = \sqrt{\frac{1}{T} \sum_{t=1}^T \min(R_t, 0)^2}$$

- ▶ $\hat{\sigma}_D$: the downside semi-deviation; days with $R_t \geq 0$ contribute 0
- ▶ Sortino > Sharpe when losses are smaller or rarer than gains

- ▣ **Calmar ratio** = CAGR/|MDD| (previous slide)

- ▶ annual growth per unit of maximum loss from a peak
- ▶ it depends on a single episode, so it is descriptive, with no simple standard error

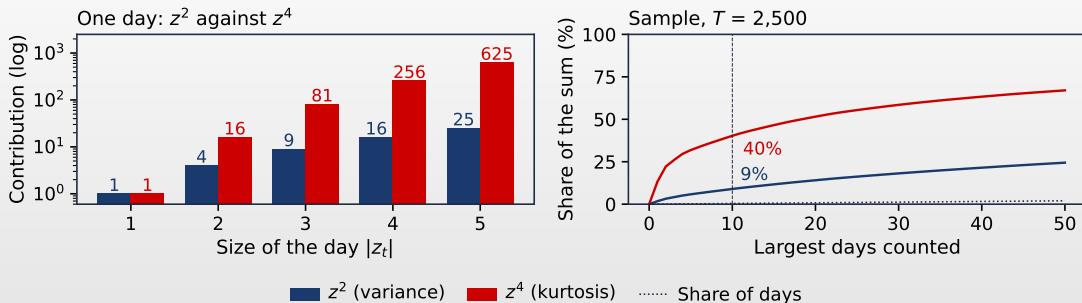
Prerequisites for Today (6/23): Moments and the Fourth Power

- **Moment of order k :** the mean of the k -th power
 - ▶ for returns: $\mathbb{E}[r^k]$; for standardised returns $z = (r - \mu)/\sigma$: $\mathbb{E}[z^k]$
 - ▶ $k = 1$: the mean; $k = 2$: the variance; $k = 3$: the skewness S ; $k = 4$: the kurtosis K
 - ▶ **finite fourth moment:** $\mathbb{E}[r^4] < \infty$; otherwise $\hat{K}$ does not settle down as T grows
- **Sample kurtosis:** the mean of the fourth powers of the standardised returns

$$\hat{K} = \frac{1}{T} \sum_{t=1}^T z_t^4, \quad z_t = \frac{r_t - \bar{r}}{\hat{\sigma}}$$

- ▶ $\hat{K} = 3$ for the Normal distribution; the **excess kurtosis** $\hat{K} - 3 > 0$ signals heavy tails
- Why the fourth power measures the tails
 - ▶ a day's contribution grows very fast with $|z_t|$: $1^4 = 1$, $2^4 = 16$, $3^4 = 81$, $10^4 = 10,000$
 - ▶ so $\hat{K}$ is driven by the extreme days, not by ordinary days
 - ▶ one day with $z_t = 10$ alone adds $10^4/T$ to $\hat{K}$: $+4$ for $T = 2,500$

Prerequisites for Today (7/23): How Extreme Days Drive the Kurtosis



- Left: the contribution of a single day to $\sum_t z_t^2$ (variance) and to $\sum_t z_t^4$ (kurtosis), on a log scale
 - ▶ a day with $|z_t| = 5$ counts 25 times as much as a day with $|z_t| = 1$ in the variance, but 625 times as much in the kurtosis
- Right: a simulated heavy-tailed sample, $T = 2,500$; the share of the largest days in the two sums
 - ▶ the 10 largest days (0.4% of days) give 9% of $\sum_t z_t^2$ but 40% of $\sum_t z_t^4$
 - ▶ this is why $\hat{K}$ moves a lot when a few days are removed (B1)

Prerequisites for Today (8/23): Quantiles and a Robust Tail Measure

- ▣ **Quantile** of order p , written q_p : the value below which a fraction p of the returns lies

$$q_p = F^{-1}(p), \quad F(x) = P(r_t \leq x)$$

- ▶ F : the distribution function; empirically, q_p is the value at position pT in the returns sorted in increasing order
- ▶ $q_{0.5}$ is the median; $q_{0.25}$ and $q_{0.75}$ are the quartiles

- ▣ Robust alternative [Kim & White \(2004\)](#): the Crow–Siddiqui quantile excess kurtosis

$$\frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$$

- ▶ numerator: the width of the range holding 95% of the days; denominator: the width of the middle 50%
- ▶ 2.91 is the value of the ratio for the Normal distribution, so the measure is 0 for the Normal distribution
- ▶ positive value: the tails are long relative to the centre, longer than for the Normal distribution
- ▶ it depends only on quantiles, so a few extreme days barely move it

Prerequisites for Today (9/23): The Jarque–Bera Test

- ▣ **Jarque–Bera** Jarque & Bera (1980): measures how far the skewness and the excess kurtosis are from their Normal values, both 0

$$JB = T \left[\frac{\hat{S}^2}{6} + \frac{(\hat{K} - 3)^2}{24} \right]$$

- ▶ 6 and 24: the variances of $\sqrt{T} \hat{S}$ and $\sqrt{T}(\hat{K} - 3)$ under H_0 , for large T
- ▶ each term is therefore the square of an approximately $N(0, 1)$ variable
- ▶ $JB \geq 0$, and $JB = 0$ only if $\hat{S} = 0$ and $\hat{K} = 3$
- ▣ H_0 : i.i.d. returns with the Normal distribution
 - ▶ under H_0 , $JB \sim \chi^2_2$ for large T (two terms); 5% critical value 5.99
 - ▶ a large JB: the skewness or the tails do not fit the Normal distribution
- ▣ Example: $T = 1\,000$, $\hat{S} = 0$, $\hat{K} = 6$: $JB = 1\,000 \times 9/24 = 375 > 5.99$, reject H_0

Prerequisites for Today (10/23): Value-at-Risk and the Normal Distribution

- ▣ **Value-at-Risk (VaR) 1%** over one day: the loss threshold exceeded with probability 1%

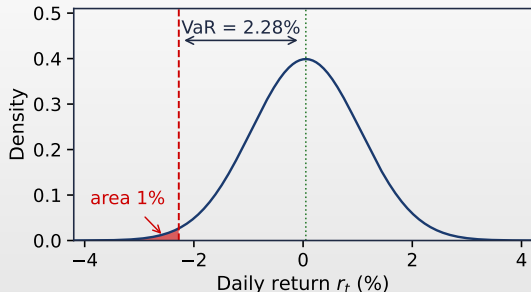
$$P(L_t > \text{VaR}_\alpha) = \alpha, \quad L_t = -r_t, \quad \text{VaR}_\alpha = -q_\alpha$$

- ▶ L_t : the loss on day t ; α : the VaR level, here $1\% = 0.01$
 - ▶ q_α : the α -quantile of the return; the minus sign turns the negative quantile into a positive loss
 - ▶ VaR is not the maximum loss: on about 1% of days the loss exceeds it, and VaR does not say by how much
- ▣ If the return has the Normal distribution $N(\mu, \sigma^2)$

$$q_\alpha = \mu + \sigma \Phi^{-1}(\alpha)$$

- ▶ $\Phi^{-1}(0.01) = -2.326$: 1% of the values of an $N(0, 1)$ variable lie below -2.326
 - ▶ example: $\mu = 0.05\%$, $\sigma = 1\%$: $q_{0.01} = 0.05 - 2.326 = -2.28\%$, so $\text{VaR } 1\% = 2.28\%$
- ▣ **Empirical VaR**: minus the sample 1% quantile
 - ▶ on the sample that produced it, it is breached on 1% of days by construction

Prerequisites for Today (11/23): The Quantile and VaR on a Density



- Density of $r_t \sim N(\mu, \sigma^2)$
- Left tail, probability $\alpha = 1\%$
- Quantile $q_{0.01} = -2.28\%$
- Mean $\mu = 0.05\%$

- **Density** f : the area under the curve between a and b is $P(a < r_t \leq b)$; the total area is 1

- The density of the Normal distribution $N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- ▶ symmetric around μ ; its width is set by σ

- **Quantile** $q_{0.01}$: the point that leaves an area of 1% to its left (the red area)

$$\int_{-\infty}^{q_{0.01}} f(x) dx = 0.01$$

- **VaR 1%** = $-q_{0.01}$: the distance from 0 to the quantile, expressed as a positive loss

- ▶ $\mu = 0.05\%$, $\sigma = 1\%$: $q_{0.01} = -2.28\%$, VaR 1% = 2.28% (the example on the previous slide)
- ▶ with heavier tails the 1% area sits further left and VaR grows (B2)

Prerequisites for Today (12/23): The Student- t Distribution

- ▣ **Student- t** with $\nu > 0$ degrees of freedom: a symmetric distribution with heavier tails than the Normal distribution
 - ▶ $P(|X| > x)$ decays like $x^{-\nu}$ as x grows: the smaller ν , the heavier the tails
 - ▶ as $\nu \rightarrow \infty$ it becomes the Normal distribution
- ▣ Moments: $\mathbb{E}|X|^p$ is finite only for $p < \nu$
 - ▶ the variance exists for $\nu > 2$ and equals $\nu/(\nu - 2)$; the kurtosis exists for $\nu > 4$
- ▣ Quantiles: $t_\nu^{-1}(\alpha)$ is the α -quantile of the standard Student- t
 - ▶ $t_3^{-1}(0.01) = -4.54$
 - ▶ rescaled to unit variance (divided by $\sqrt{3/(3-2)} = \sqrt{3}$): $-4.54/\sqrt{3} = -2.62$, deeper than -2.33 , the quantile of the Normal distribution
- ▣ Fitting: by maximum likelihood, for the variable $m + sY$ with Y a standard Student- t
 - ▶ m : the location (centre); $s > 0$: the scale (width); `scipy.stats.t.fit`

Prerequisites for Today (13/23): The Student- t Density and the Tail Index

- ▣ **Standard Student- t density** with ν degrees of freedom

$$f_{\nu}(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

- ▶ Γ : the gamma function, $\Gamma(n) = (n-1)!$ for integer n ; the leading fraction makes the total area 1
- ▶ for large $|x|$, $f_{\nu}(x)$ decays like $|x|^{-(\nu+1)}$, a power of $|x|$
- ▶ the Normal density decays like $e^{-x^2/2}$, much faster: hence the difference in the tails

- ▣ **Tail index α** : how fast the probability of a large move falls

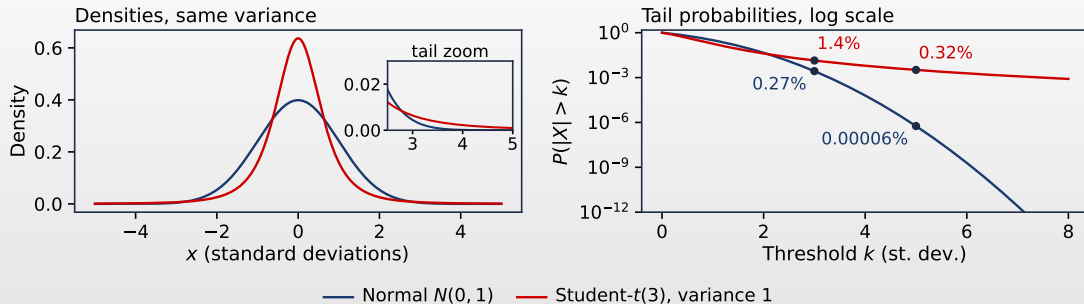
$$P(|r_t| > x) \propto x^{-\alpha}, \quad x \text{ large}$$

- ▶ α : proportional to; a small α means heavy tails; for the Student- t , $\alpha = \nu$
- ▶ $\mathbb{E}|r_t|^p$ is finite only for $p < \alpha$: with $\alpha \leq 4$ the kurtosis does not exist

- ▣ After the lecture

- ▶ **A5** and **B6** estimate α with the Hill estimator; **A6** derives the variance of $\hat{\rho}_k$

Prerequisites for Today (14/23): The Normal and the Student- t with the Same Variance



- Left: $N(0, 1)$ and the Student- t with $\nu = 3$, rescaled to unit variance (divided by $\sqrt{3}$)
 - ▶ same variance, but the Student- t has a higher peak, thinner shoulders and heavier tails (zoom)
- Right: $P(|X| > k)$ on a log scale
 - ▶ $k = 3$: 1.4% for the Student- t against 0.27% for the Normal distribution, 5 times as likely
 - ▶ $k = 5$: 0.32% against 0.00006%, more than 5,000 times as likely

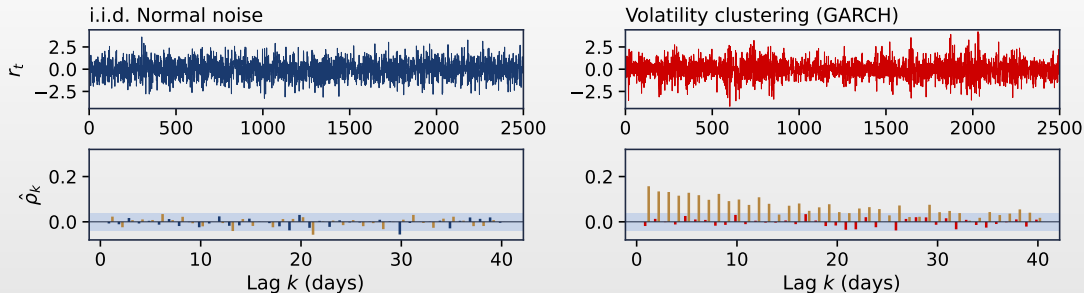
Prerequisites for Today (15/23): The ACF and the Ljung–Box Test

- **ACF** (autocorrelation function): the sequence $\hat{\rho}_k$, $k = 1, 2, \dots$, plotted against the lag
 - ▶ for i.i.d. returns $\hat{\rho}_k \approx N(0, 1/T)$, so the standard deviation of $\hat{\rho}_k$ is $1/\sqrt{T}$
 - ▶ 95% band: $\pm 1.96/\sqrt{T}$ ($T = 2500$: ± 0.039); a $\hat{\rho}_k$ outside it is significant at 5%
- **Volatility clustering** in the ACF: $|r_t|$ and r_t^2 are autocorrelated although r_t is not
- **Ljung–Box** [Ljung & Box \(1978\)](#): tests the first m autocorrelations jointly

$$Q(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$$

- ▶ m : the number of lags tested; $H_0: \rho_1 = \dots = \rho_m = 0$
- ▶ for i.i.d. returns $Q(m) \sim \chi_m^2$; a Q above the critical value: at least one autocorrelation is non-zero
- ▶ on r_t it tests linear predictability; on r_t^2 it tests volatility clustering

Prerequisites for Today (16/23): The ACF of i.i.d. Noise and under Volatility Clustering



— ACF of r_t (i.i.d.) — ACF of r_t (GARCH) — ACF of $|r_t|$ band $\pm 1.96/\sqrt{T}$

- The simulated series on the right: $r_t = \sigma_t \varepsilon_t$, $\sigma_t^2 = 0.05 + 0.10 r_{t-1}^2 + 0.85 \sigma_{t-1}^2$, ε_t i.i.d. $N(0, 1)$
 - ▶ a day with a large $|r_{t-1}|$ raises today's volatility: calm and turbulent periods alternate
- The ACF of r_t stays inside the band $\pm 1.96/\sqrt{T} = \pm 0.039$ ($T = 2,500$) for both series
- The ACF of $|r_t|$: inside the band for i.i.d. noise, far above it for the series on the right (0.16 at lag 1)

Prerequisites for Today (17/23): The ARCH-LM Test

- ▣ **ARCH-LM Engle (1982)**: tests whether today's squared return depends on past squared returns, i.e. volatility clustering
- ▣ The auxiliary regression, estimated by OLS

$$x_t^2 = a_0 + a_1 x_{t-1}^2 + \cdots + a_q x_{t-q}^2 + u_t$$

- ▶ q : the number of lags; $a_0, \dots, a_q$: the coefficients; u_t : the error
- ▶ $H_0: a_1 = \cdots = a_q = 0$, i.e. no volatility clustering
- ▣ The statistic $LM = (T - q)R^2$, compared with χ_q^2
 - ▶ $R^2 \in [0, 1]$: the share of the variation of x_t^2 explained by past squares
 - ▶ $T - q$: the number of observations used in the regression
 - ▶ example: $T = 1000$, $q = 5$, $R^2 = 0.02$: $LM = 995 \times 0.02 = 19.9 > 11.07$, the 5% critical value of χ_5^2 : clustering

Prerequisites for Today (18/23): The Mixture of Normal Distributions

- ▣ **Mixture of Normal distributions** Clark (1973): the return is a Normal shock scaled by a random volatility

$$r_t = \sigma_t \varepsilon_t$$

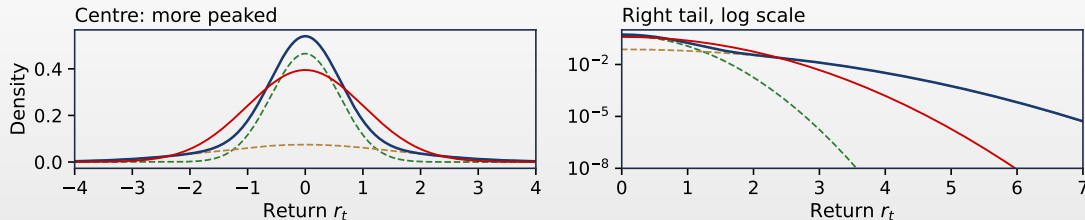
- ▶ ε_t : the shock, i.i.d. $N(0, 1)$ and independent of σ_t
- ▶ $\sigma_t > 0$: the volatility of day t , itself random (for example, driven by the pace of information arrival)

- ▣ The population kurtosis of r_t

$$K = \frac{\mathbb{E}r^4}{(\mathbb{E}r^2)^2} = \frac{3\mathbb{E}\sigma^4}{(\mathbb{E}\sigma^2)^2} \geq 3$$

- ▶ 3 is $\mathbb{E}\varepsilon^4$, the kurtosis of $N(0, 1)$
- ▶ $K = 3$ only if σ_t is constant; the more σ_t varies, the larger K
- ▶ heavy tails from Normal shocks (A4)

Prerequisites for Today (19/23): A Mixture of Two Normal Distributions



- calm days: $0.7 \times N(0, 0.6^2)$
- turbulent days: $0.3 \times N(0, 1.6^2)$
- Mixture (kurtosis 5.9)
- Normal, same variance 1.02 (kurtosis 3)

□ Calm days, with probability 0.7: $\sigma_t = 0.6$; turbulent days, with probability 0.3: $\sigma_t = 1.6$

$$f(x) = 0.7 \varphi_{0.6}(x) + 0.3 \varphi_{1.6}(x), \quad \varphi_s(x) = \frac{1}{s} \varphi\left(\frac{x}{s}\right)$$

- ▶ φ_s : the density of $N(0, s^2)$; mixture variance $0.7 \times 0.6^2 + 0.3 \times 1.6^2 = 1.02$; $K = 3 \mathbb{E}\sigma^4 / (\mathbb{E}\sigma^2)^2 = 5.9$
- ▶ more very small and more very large days than under the Normal distribution with the same variance
- ▶ $P(|r_t| > 4)$: 0.37% in the mixture against 0.0075% for the Normal distribution

Prerequisites for Today (20/23): Inference under Volatility Clustering

- Under volatility clustering, $\text{Var}(\hat{\rho}_k) > 1/T$
 - ▶ the band $\pm 1.96/\sqrt{T}$ is too narrow and $Q(m)$ rejects H_0 too often
- The variance of $\hat{\rho}_k$ is estimated directly from the data Lobato, Nankervis & Savin (2001)

$$\hat{\tau}_k = \frac{\sum_{t>k} x_t^2 x_{t-k}^2}{(\sum_t x_t^2)^2}$$

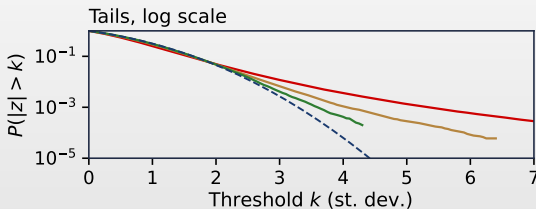
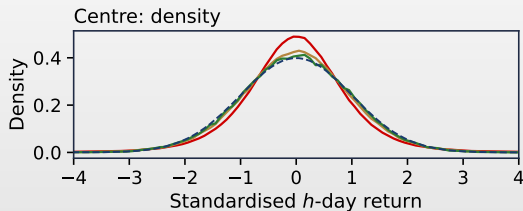
- ▶ for i.i.d. returns $\hat{\tau}_k \approx 1/T$; $T\hat{\tau}_k$ says how many times the variance of $\hat{\rho}_k$ exceeds $1/T$
- ▶ robust band $\pm 1.96\sqrt{\hat{\tau}_k}$
- Robust Ljung–Box (B4): each $\hat{\rho}_k^2$ is divided by its own estimated variance

$$\tilde{Q}(m) = \sum_{k=1}^m \frac{\hat{\rho}_k^2}{\hat{\tau}_k}$$

- ▶ the same H_0 as $Q(m)$, still compared with χ_m^2
- ▶ example: $T\hat{\tau}_1 = 4$ makes the robust band $\sqrt{4} = 2$ times wider than $\pm 1.96/\sqrt{T}$

Prerequisites for Today (21/23): Aggregational Gaussianity

- **The h -day log return:** the sum of h consecutive daily returns, $r_t^{(h)} = r_t + r_{t-1} + \dots + r_{t-h+1}$
 - ▶ h : the horizon, in days; κ : the excess kurtosis of the daily return
 - ▶ for i.i.d. returns the h -day excess kurtosis is $\kappa/h \rightarrow 0$: the sum approaches the Normal distribution



— $h = 1$: excess kurtosis 6.00 — $h = 20$: excess kurtosis 0.30
 — $h = 5$: excess kurtosis 1.20 - - - Normal $N(0, 1)$

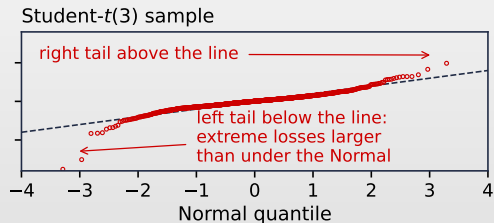
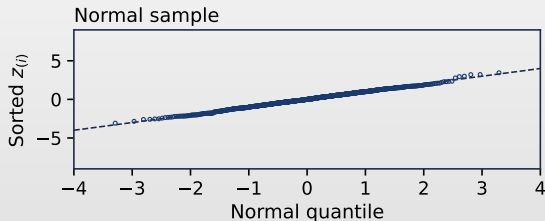
- Simulated i.i.d. heavy-tailed daily returns ($\kappa = 6$), summed over $h = 1, 5, 20$ days and standardised
 - ▶ the excess kurtosis falls from 6 to 1.2 and to 0.3; the tail (right) approaches that of the Normal distribution
 - ▶ under volatility clustering the approach is slower than κ/h (B5)

Prerequisites for Today (22/23): The QQ Plot

- **QQ plot:** the data quantiles against the quantiles of $N(0, 1)$, one point per day

$$\left(\Phi^{-1}\left(\frac{i - 0.5}{T}\right), z_{(i)} \right), \quad i = 1, \dots, T$$

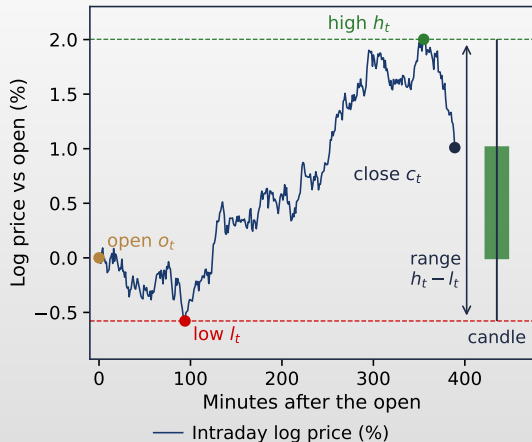
- $z_{(1)} \leq \dots \leq z_{(T)}$: the standardised returns sorted in increasing order; $(i - 0.5)/T$: the matching proportion



— 45-degree line: perfect Normal fit ○ Normal: points on the line ○ Heavy tails: S-shape

- On the 45-degree line: the Normal distribution (left); an S shape: heavy tails (right, Student- t with $\nu = 3$)
 - the left tail lies below the line and the right tail above it: the extreme data quantiles exceed the Normal quantiles

Prerequisites for Today (23/23): The Daily Range and Parkinson Volatility



- For the extension packs (A8, B15)
- O_t, H_t, L_t, C_t : the open, high, low and close of day t
 - ▶ in lower case, their logs: $o_t = \ln O_t$, $h_t = \ln H_t$, $l_t = \ln L_t$, $c_t = \ln C_t$
- **Log range** $h_t - l_t = \ln(H_t/L_t)$: the amplitude of the day
 - ▶ it uses the high and the low of the path, not only the closing price
 - ▶ in the chart: the range is 2.58%, while $c_t - o_t$ is only 1.01%
- **Parkinson estimator** Parkinson (1980)

$$\hat{\sigma}_P^2 = \frac{1}{4 \ln 2} \cdot \frac{1}{n} \sum_{t=1}^n (h_t - l_t)^2$$

- ▶ n : the number of days; $\hat{\sigma}_P^2$: the estimated daily variance
- ▶ $4 \ln 2 = 2.77$: for a driftless Brownian motion, $\mathbb{E}[(h_t - l_t)^2] = 4 \ln 2 \sigma^2$

A1 [Solved]: Returns and Annualisation — Task

▣ Questions

- ▶ which return adds up over time?
- ▶ how do daily numbers become annual ones?

▣ Inputs

- ▶ a share closes at 100, 110, 99, 104.5 on days 0–3 and pays no dividend
- ▶ an index has a daily mean log return of 0.04% and a daily volatility of 1.2%
- ▶ Bitcoin has a daily volatility of 3.5%
- ▶ returns are serially uncorrelated with constant variance

▣ Tasks

1. Compute the three simple returns R_t and the three log returns r_t , all in %.
2. Compute the 3-day returns $P_3/P_0 - 1$ and $\ln(P_3/P_0)$ and compare them with $\sum_t R_t$ and $\sum_t r_t$.
3. Annualise the index mean log return and volatility with $A = 252$.
4. Annualise the Bitcoin volatility and justify the choice of A .

▣ **Report:** a table day, price, R_t , r_t and the three annual numbers

▣ Notebook: section [A1](#)

A1 [Solved]: Solution (1/2)



A1 [Solved]: Solution (2/2)

t	P_t	R_t	r_t
0	100	—	—
1	110	+10.00%	+9.53%
2	99	−10.00%	−10.54%
3	104.5	+5.56%	+5.41%
sum		5.56%	4.40%
0 → 3		4.50%	4.40%

- The sum of the log returns is $\ln(104.5/100)$: the intermediate terms $\ln P_t$ cancel
- Simple returns compound:
 $1.1 \times 0.9 \times 1.0556 - 1 = 4.5\%$

□ Index

- ▶ annualised mean log return $252 \times 0.04\% = 10.08\%$
- ▶ volatility $\sqrt{252} \times 1.2\% = 15.87 \times 1.2\% = 19.05\%$

□ Bitcoin trades every day: $\sqrt{365} \times 3.5\% = 66.9\%$

- ▶ with $\sqrt{252}$ one reports 55.6%, 17% too low

A2(a) [Proposed]: The Depth of a Drawdown — Task

▣ **Model:** A1 [Solved]; notebook section A2

▣ **Questions**

- ▶ what is the worst loss from a previous peak?
- ▶ how does it compare with the growth rate?

▣ **Inputs**

- ▶ year-end portfolio values $V_0, \dots, V_5 = 100, 120, 90, 110, 130, 117$
- ▶ 5-year CAGR $(V_5/V_0)^{1/5} - 1 = 3.19\%$: the constant annual rate that takes V_0 to V_5

▣ **Tasks:** complete the worksheet

1. Compute the running peak $M_t = \max_{s \leq t} V_s$, the largest V_s over years $s = 0, \dots, t$.
2. Compute the drawdown $DD_t = V_t/M_t - 1 \leq 0$, the relative loss from the peak.
3. Find the maximum drawdown $MDD = \min_t DD_t$ and the year in which it occurs.
4. Compute the Calmar ratio $\text{CAGR}/|MDD|$, the annual growth per unit of maximum loss.

Year	0	1	2	3	4	5
V_t	100	120	90	110	130	117
M_t	100	120	...	...	...	...
DD_t	0%	0%	...	...	...	...

▣ **Interpretation:** why is the fall to 117 in year 5 not the maximum drawdown?

B1 [Solved]: Descriptive Statistics — Task

▣ Questions

- ▶ how heavy are the tails of BET-TR, Bitcoin and EUR/RON?
- ▶ how much does the kurtosis depend on a few days?

▣ Data: r_{bet} , r_{btc} , r_{eur} from the setup

- ▶ $T = 2\,992, 4\,384, 5\,352$; $A = 252, 365, 252$

▣ Tasks

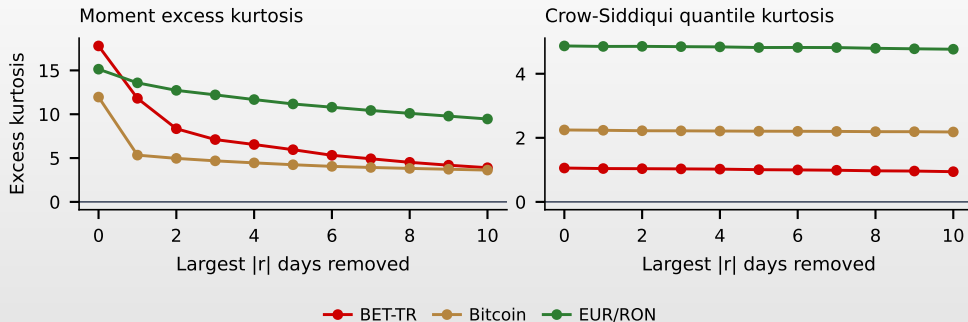
1. Compute the mean, the standard deviation, the annualised volatility, the skewness, the excess kurtosis, the minimum and the maximum of each series.
 2. Compute $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$ and its two terms.
 3. Compare JB with 5.99, the 5% critical value of χ^2_2 .
 4. Recompute each excess kurtosis after removing the day with the largest $|r_t|$, and after removing the five largest.
 5. Recompute the Crow–Siddiqui quantile excess kurtosis $\frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$ on the same trimmed samples (q_p : the quantile of order p ; 0 for the Normal distribution).
- ▣ **Report:** one table for steps 1–3 and a chart of kurtosis against the number of days removed
- ▣ **Interpretation:** which measure of tail heaviness can you report with confidence?

B1 [Solved]: Solution (1/2) — BET-TR Step by Step

- Step 1: $\bar{r} = 0.077\%$, $\hat{\sigma} = 0.978\%$, volatility $\sqrt{252} \times 0.978\% = 15.5\%$
 - ▶ $\hat{S} = -1.40$, $\hat{K} - 3 = 17.79$
 - ▶ worst day -11.86% on 19 Dec 2018 (the bank-asset tax)
 - ▶ best day $+6.71\%$ on 24 Dec 2018
- Step 2: $JB = 40\,433$
 - ▶ skewness term $2\,992 \times 1.40^2/6 = 977$
 - ▶ kurtosis term $2\,992 \times 17.79^2/24 = 39\,455$, 98% of the statistic
- Step 3: $40\,433 > 5.99$, $p < 10^{-10}$
- Step 4: excess kurtosis 17.8 with all days
 - ▶ 11.8 without 19 Dec 2018
 - ▶ 5.95 without the five largest days: 19 Dec 2018, 16 Mar 2020, 9 Mar 2020, 24 Dec 2018, 9 Mar 2022
- Step 5: Crow–Siddiqui 1.06 with all days, 1.01 without the five largest days
- One day carries a third of the moment kurtosis
 - ▶ the quantile measure does not move

	BET-TR	Bitcoin	EUR/RON
Mean (% a day)	0.077	0.118	0.007
Std. dev. (% a day)	0.978	3.485	0.279
Volatility (% a year)	15.5	66.6	4.4
Skewness	-1.40	-0.70	0.73
Excess kurtosis	17.8	12.0	15.1
Min (%)	-11.9	-46.5	-2.5
Max (%)	6.7	22.5	2.9
JB	40 433	26 482	51 575
Top 5 removed	5.95	4.24	11.17

B1 [Solved]: Solution (2/2) — Sensitivity



- All JB statistics exceed the critical value 5.99 more than 4 000-fold
- Bitcoin without 12 Mar 2020 alone: kurtosis 12.0 → 5.3
- Interpretation: the tails are heavy by both measures
 - ▶ report the quantile excess kurtosis: its value does not hinge on a few days

A3 [Solved]: Jarque–Bera and Ljung–Box, Step by Step — Task

▣ Questions

- ▶ how does a statistic become a decision?
- ▶ which assumptions does the decision need?

▣ Inputs: two hypothetical samples of daily returns; tests at the 5% level

- ▶ sample 1: $T = 2\,500$, skewness $\hat{S} = -0.5$, kurtosis $\hat{K} = 8$ (not excess)
- ▶ sample 2: $T = 1\,000$, $\hat{\rho}_1 = 0.10$, $\hat{\rho}_2 = -0.05$, $\hat{\rho}_3 = 0.03$

▣ Tasks

1. Compute the two terms of $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$ Jarque & Bera (1980), the statistic, and $p = P(\chi_2^2 \geq JB)$.
2. Compute the three terms of $Q(3) = T(T+2) \sum_{k=1}^3 \hat{\rho}_k^2 / (T-k)$ Ljung & Box (1978), the statistic, and $p = P(\chi_3^2 \geq Q)$.
3. State the H_0 of each test.
4. Name the assumption of each χ^2 reference distribution that fails under volatility clustering.

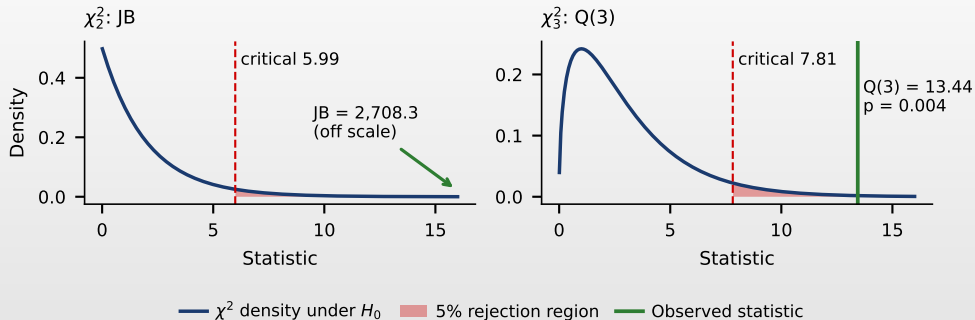
▣ Report: the summands, the two statistics, the p-values and the decisions

▣ Notebook: section A3

A3 [Solved]: Solution (1/2) — Step by Step

1. $\hat{S}^2/6 = 0.25/6 = 0.042$; $(\hat{K} - 3)^2/24 = 25/24 = 1.042$
 - ▶ $JB = 2500 \times 1.084 = 2708.3$, $p < 10^{-10}$: reject
 - ▶ 96% of JB is the kurtosis term
2. $0.01/999 = 1.001 \times 10^{-5}$, $0.0025/998 = 0.251 \times 10^{-5}$, $0.0009/997 = 0.090 \times 10^{-5}$
 - ▶ $Q(3) = 1000 \times 1002 \times 1.342 \times 10^{-5} = 13.44$, $p = 0.004$: reject
3. H_0 of each test
 - ▶ JB: i.i.d. returns with the Normal distribution
 - ▶ Ljung–Box: no autocorrelation
4. The χ^2_3 reference of Ljung–Box also needs i.i.d. returns
 - ▶ under clustering $\text{Var}(\hat{\rho}_k) > 1/T$ and Q rejects too often (A6, B4)

A3 [Solved]: Solution (2/2) — Reference Distributions



- The p-value is the area of the χ^2 density beyond the observed statistic
- Interpretation: for an i.i.d. sample both rejections are sound
 - ▶ for clustered returns only the JB rejection of "i.i.d. and Normal" is sound
 - ▶ the $Q(3)$ p-value is then too small

B2 [Proposed]: Heavy Tails and VaR 1% on BET-TR — Task (1/2)

- **Model:** B1 [Solved]; notebook section B2
- **Question:** how large is the one-day VaR 1%, the loss threshold exceeded with probability 1%, under three distributions?
- **Data:** r_{bet} , $T = 2992$
 - ▶ models fitted and checked on the **same** sample: a descriptive comparison
 - ▶ not a backtest (a check of VaR on data after the estimation window)
- **Tasks**
 1. Standardise $z_t = (r_t - \bar{r})/\hat{\sigma}$ and count the days with $|z_t| > 3$.
 2. Compare their share with the share under the Normal distribution, $P(|Z| > 3) = 0.27\%$ for $Z \sim N(0, 1)$.
 3. Fit a Student- t to z_t by maximum likelihood (`scipy.stats.t.fit`: degrees of freedom ν , location m , scale s).

B2 [Proposed]: Heavy Tails and VaR 1% on BET-TR — Task (2/2)

▣ **Tasks** (continued)

4. Compute $\text{VaR}_{1\%} = -q_r$, with $q_r = \bar{r} + \hat{\sigma}q_z$, for three choices of the 1% quantile.
 - q_z : the 1% quantile of z_t ; q_r : the same quantile in return units (undoing the standardisation)
 - Student- t : $q_z = m + s t_{\nu}^{-1}(0.01)$, with t_{ν}^{-1} the standard Student- t quantile
 - Normal distribution: $q_z = \Phi^{-1}(0.01) = -2.326$
 - empirical: q_r = the 1% quantile of r_t
 5. Count the breaches (days with $r_t < q_r$, i.e. a loss larger than the VaR) and compare them with the expected count $0.01 T = 29.9$.
- ▣ **Report**: a table method / VaR / breaches and a lower-tail plot with the three quantiles
- ▣ **Interpretation**: why does the empirical VaR have about 30 breaches by construction?

A4 [Solved]: Kurtosis of a Volatility Mixture — Task

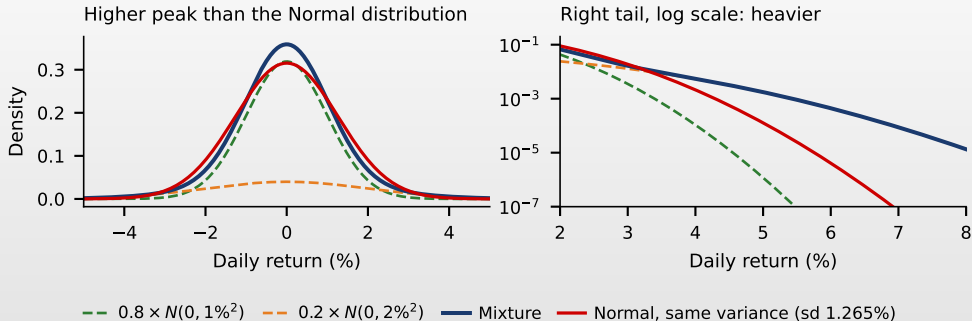
- ▣ **Question:** can random volatility alone create heavy tails?
- ▣ **Assumptions:** $r_t = \sigma_t \varepsilon_t$, ε_t i.i.d. $N(0, 1)$ and independent of σ_t
 - ▶ σ_t : the random volatility of day t ; ε_t : the standardised shock
 - ▶ $\sigma_t = 1\%$ with probability 0.8 and 2% with probability 0.2
 - ▶ σ_t is drawn **independently every day**
- ▣ **Tasks**
 1. Using independence and $\mathbb{E}\varepsilon^4 = 3$, show that the population kurtosis (not the excess) is $K = \mathbb{E}r^4 / (\mathbb{E}r^2)^2 = 3 \mathbb{E}\sigma^4 / (\mathbb{E}\sigma^2)^2$.
 2. Complete a two-state table with the probability, σ^2 (in $\%^2$) and σ^4 (in $\%^4$).
 3. Compute $\mathbb{E}\sigma^2$, $\mathbb{E}\sigma^4$ and K .
 4. Use Jensen's inequality, $\mathbb{E}[X^2] \geq (\mathbb{E}X)^2$ with $X = \sigma^2$, to show that $K \geq 3$.
 5. State whether this model produces volatility clustering.
- ▣ **Report:** the table, K and the excess kurtosis
- ▣ **Interpretation:** link the result to the mixture-of-distributions hypothesis of [Clark \(1973\)](#)
- ▣ Notebook: section [A4](#)

A4 [Solved]: Solution (1/2) — Moments

State	Prob.	σ^2	σ^4
calm	0.8	1	1
turbulent	0.2	4	16
mean		1.6	4.0

1. $\mathbb{E}r^4 = \mathbb{E}\sigma^4 \mathbb{E}\varepsilon^4 = 3\mathbb{E}\sigma^4$ and $\mathbb{E}r^2 = \mathbb{E}\sigma^2$
2. The table on the left
3. $\mathbb{E}\sigma^2 = 1.6$, $\mathbb{E}\sigma^4 = 4.0$
 - $K = 3 \times 4.0 / 1.6^2 = 12 / 2.56 = 4.69$; excess kurtosis 1.69
4. $\mathbb{E}[(\sigma^2)^2] \geq (\mathbb{E}\sigma^2)^2$, with equality only for a constant σ : $K \geq 3$
5. No clustering: the states are independent across days
 - so $|r_t|$ is uncorrelated
 - A9 makes the states persistent

A4 [Solved]: Solution (2/2) — Densities

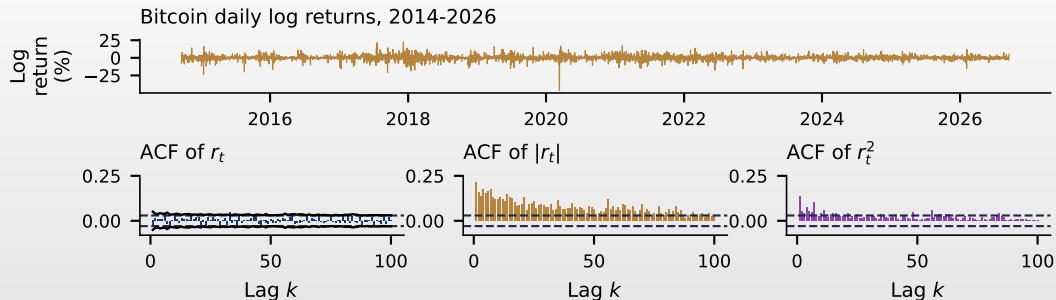


- $P(|r_t| > 4\%)$: 0.92% in the mixture, 0.16% for the Normal distribution with the same variance
- Interpretation: the random operational time of [Clark \(1973\)](#) (the pace of information arrival) is such a mixture
 - variation in volatility creates heavy tails

B3 [Solved]: Uncorrelated but Not Independent (Bitcoin) — Task

- ▣ **Question:** do past Bitcoin returns predict the direction or only the size of today's return?
- ▣ **Data:** `r_btc`, 18 Sep 2014 – 18 Sep 2026, $T = 4\,384$
- ▣ **Tasks**
 1. Plot the returns.
 2. Plot the ACF of r_t , $|r_t|$ and r_t^2 for lags 1–100 with the i.i.d. reference band $\pm 1.96/\sqrt{T} = \pm 0.030$.
 3. Compute the Ljung–Box $Q(10)$ on r_t and on r_t^2 (`acorr_ljungbox`), with p-values from χ_{10}^2 .
 4. Run the ARCH-LM test with 5 lags [Engle \(1982\)](#).
 - regress x_t^2 on a constant and $x_{t-1}^2, \dots, x_{t-5}^2$
 - compare $(T - 5)R^2$ with χ_5^2
- ▣ **Report:** the chart, the lag-1 autocorrelations and the three statistics
 - ▶ label each with its series and its H_0
- ▣ **Interpretation:** the $Q(10)$ on r_t has $p \approx 3\%$
 - ▶ given step 4, is its i.i.d. reference distribution valid?
- ▣ Notebook: section [B3](#)

B3 [Solved]: Solution (1/2) — Returns and ACF



-- i.i.d. reference band $\pm 1.96/\sqrt{T}$ (valid only under independence) — Robust band for r_t , $\pm 1.96\sqrt{\hat{\tau}_k}$

- Direction: no structure in the ACF of r_t
- Size: $|r_t|$ and r_t^2 autocorrelated for dozens of lags

B3 [Solved]: Solution (2/2) — Tests

2. $\hat{\rho}_1$: -0.022 for r_t (inside the band), 0.213 for $|r_t|$ and 0.137 for r_t^2
3. $Q(10)$ results
 - ▶ on r_t : 20.2 , $p = 0.028$
 - ▶ on r_t^2 : 212.7 , $p \approx 4 \times 10^{-40}$
4. $\text{ARCH-LM}(5) = 114.7$, $p \approx 4 \times 10^{-23}$
 - ▶ its χ^2 reference needs $\mathbb{E}r^4 < \infty$, so the p-value is indicative
- The sizes of returns are dependent: returns are not i.i.d.
 - ▶ the 2.8% p-value of $Q(10)$ on r_t rests on a false assumption
- The i.i.d. band is a reference, not a valid test for $|r_t|$ and r_t^2
 - ▶ A6 and B4 correct the inference for r_t

B4 [Solved]: Heteroskedasticity-Robust Ljung–Box — Task (1/2)

- ▣ **Question:** which autocorrelations of returns survive once the variance of $\hat{\rho}_k$ allows for clustering?
- ▣ **Data:** `r_sp`, `r_bet`, `r_eur`, `r_btc`; $x_t = r_t - \bar{r}$
- ▣ **Tasks**
 1. For $k = 1, \dots, 20$ compute $\hat{\rho}_k$ and $\hat{\tau}_k = \sum_{t=k+1}^T x_t^2 x_{t-k}^2 / (\sum_{t=1}^T x_t^2)^2$ Lobato, Nankervis & Savin (2001), the estimated variance of $\hat{\rho}_k$.
 2. Plot $\hat{\rho}_k$ with the classical band $\pm 1.96/\sqrt{T}$ and the robust band $\pm 1.96\sqrt{\hat{\tau}_k}$.
 3. Compute the classical $Q(10)$ and the robust $\tilde{Q}(10) = \sum_{k=1}^{10} \hat{\rho}_k^2 / \hat{\tau}_k$.
 4. Compare both with χ_{10}^2 , whose 5% critical value is 18.3.
 5. Count the lags among 1–20 outside each band.
 - these are pointwise checks, distinct from the joint 10-lag test of steps 3–4

B4 [Solved]: Heteroskedasticity-Robust Ljung–Box — Task (2/2)

▣ Assumptions of $\tilde{Q}$

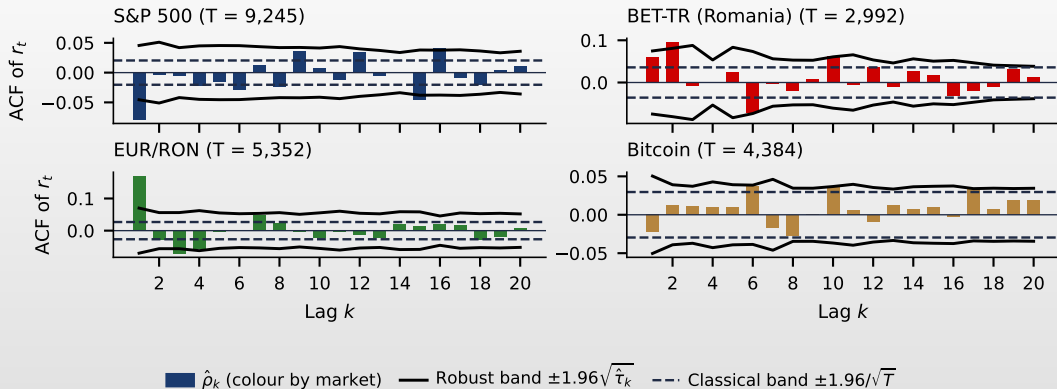
- ▶ a finite fourth moment, $\mathbb{E}r^4 < \infty$, needed for $\hat{\tau}_k$ to estimate a finite variance
- ▶ zero cross terms, $\mathbb{E}[x_t^2 x_{t-k} x_{t-l}] = 0$ for $k \neq l$
- ▶ i.e. x_t^2 is uncorrelated with the products $x_{t-k} x_{t-l}$ of past returns at different lags

▣ Interpretation

- ▶ for which series does autocorrelation survive?
- ▶ is the fourth-moment assumption credible?

▣ Notebook: section B4

B4 [Solved]: Solution (1/4) — ACF with Both Bands



B4 [Solved]: Solution (2/4) — ACF with Both Bands

- S&P 500, lag 1: $\hat{\rho}_1 = -0.079$, $T\hat{\rho}_1 = 5.00$
 - ▶ the variance of $\hat{\rho}_1$ is 5 times the i.i.d. value $1/T$
 - ▶ the robust band is $1.96\sqrt{5.00/9245} = 0.046$ instead of 0.020
- At lag 1 the robust band is $\sqrt{T\hat{\rho}_1} = 1.7$ (Bitcoin) to 2.6 (EUR/RON) times wider than the classical one

 MFM_ch1_seminar_tests

B4 [Solved]: Solution (3/4) — Joint Tests and Interpretation

	S&P 500	BET-TR	EUR/RON	Bitcoin
$Q(10)$ (p)	91.6 ($< 10^{-10}$)	70.0 ($< 10^{-10}$)	221.7 ($< 10^{-10}$)	20.2 (0.028)
$\tilde{Q}(10)$ (p)	19.0 (0.041)	16.9 (0.076)	37.4 (< 0.001)	11.9 (0.29)
Lags outside, classical / robust	8 / 3	5 / 2	6 / 2	3 / 0
Lags outside the robust band	1, 15, 16	2, 10	1, 3	none

- At 5% the classical test rejects for all four series
 - ▶ the robust test rejects for the S&P 500 (barely) and EUR/RON only
- Bitcoin (B3): the 2.8% p-value becomes 29%
 - ▶ no evidence of linear autocorrelation at lags 1–10
- EUR/RON keeps $\hat{\rho}_1 = 0.17$
 - ▶ under the managed float, the BNR reference rate adjusts in small steps over several days

B4 [Solved]: Solution (4/4) — Remaining Assumptions

- Zero cross terms
 - ▶ the test of [Dalla, Giraitis & Phillips \(2022\)](#) keeps only the significant cross terms
 - ▶ it finds none for the S&P 500, BET-TR and Bitcoin
- Finite fourth moment
 - ▶ if $P(|r_t| > x)$ decays like $x^{-\nu}$ with $\nu \leq 4$, $\mathbb{E}r^4$ is infinite
 - ▶ the robust p-values are then not valid
 - ▶ [A5](#) and [B6](#) estimate this exponent after the lecture

B5 [Proposed]: Aggregational Gaussianity (Bitcoin) — Task

- ▣ **Model:** B1 and B3 [Solved]; notebook section B5
- ▣ **Question:** do h -day Bitcoin returns approach the Normal distribution as h grows?
- ▣ **Data:** r_btc , $T = 4\,384$, in non-overlapping blocks from the first day
 - ▶ the first 5-day block is 18–22 Sep 2014, the second 23–27 Sep 2014
 - ▶ the incomplete last block (4 days for $h = 5$ and $h = 20$) is dropped
- ▣ **Tasks**
 1. Build the h -day log returns (sums of h consecutive daily returns) for $h = 1, 5, 20$.
 2. Draw a QQ plot of each standardised series against the Normal distribution, with N , the number of blocks, in the title.
 3. Compute the excess kurtosis and the i.i.d.-Gaussian reference SE $\sqrt{24/N}$.
 - $\sqrt{24/N}$: the standard error of the excess kurtosis estimated from N i.i.d. Normal observations (24, as in JB)
- ▣ **Report:** the three QQ plots and a table h , N , excess kurtosis, $\sqrt{24/N}$
- ▣ **Interpretation**
 - ▶ why is $\sqrt{24/N}$ too optimistic here?
 - ▶ why do non-overlapping blocks remove shared days but not volatility dependence?

C1 [Proposed]: Critique an AI Answer — the Answer

- **Prompt** sent to an AI assistant: “Are the daily BET-TR log returns Normally distributed? Are they autocorrelated? Give Python code and a conclusion”
- **Answer:** the code below and the conclusion

```
from scipy import stats
from statsmodels.stats.diagnostic import acorr_ljungbox
r = np.log(bet["close"]).diff().dropna()      # daily log returns
k = stats.kurtosis(r)                          # sample kurtosis
print("excess_kurtosis:", k - 3)              # the Normal has kurtosis 3
jb, p = stats.jarque_bera(r)
lb = acorr_ljungbox(r**2, lags=[10])          # Ljung-Box, 10 lags
```

- **Conclusion of the AI**
 - ▶ “Excess kurtosis 14.8: heavy tails”
 - ▶ “JB p -value 0.000: the probability that BET-TR returns follow the Normal distribution is 0%”
 - ▶ “Ljung-Box $Q(10) = 713.5$, $p < 0.001$: returns are strongly autocorrelated, so BET-TR is predictable and not a random walk”

C1 [Proposed]: Critique an AI Answer — Task (1/2)

- ▣ **Model:** A3, B3 and B4 [Solved]; notebook section C1
- ▣ **Data:** `r_bet`, the 2992 returns of the setup
- ▣ **Tasks**
 1. Find every substantive error in the code and in the three conclusions.
 2. For each error, say how you would detect it.
 - a test on simulated data or a number from the primer slides
 - a solved exercise or the library documentation
 3. Write corrected, executable code.

C1 [Proposed]: Critique an AI Answer — Task (2/2)

▣ **Tasks** (continued)

4. Report the excess kurtosis, JB with its p-value, $Q(10)$ on r_t and on r_t^2 , and the robust $\tilde{Q}(10)$ of B4.
5. State the assumptions behind the robust p-value.
6. Write the corrected conclusion in three sentences.

▣ **Report**

- ▶ the corrected code
 - ▶ a table AI value / correct value
 - ▶ a chart of the ACF of r_t and of r_t^2
- ▣ AI code usually runs without an error message
- ▶ the mistakes are in the statistics, not in the syntax
 - ▶ log what you find in `AI_ERRORS.md`

A5 [Solved]: The Hill Estimator, Step by Step — Task (1/2)

▣ Questions

- ▶ how is a tail index estimated from the largest observations?
- ▶ how precise is it?

- ▣ **Inputs:** the 11 largest absolute daily returns (%) of a hypothetical sample: 12.0; 9.5; 8.2; 7.4; 6.9; 6.1; 5.8; 5.5; 5.2; 5.0; 4.8

▣ Notation

- ▶ $X_{(1)} \geq X_{(2)} \geq \dots$: the absolute values sorted in decreasing order (the order statistics)
- ▶ k : the number of tail observations; $X_{(k+1)}$: the threshold where the tail starts

- ▣ **Assumption:** above the threshold $X_{(k+1)}$ the tail is Pareto, $P(X > x) \propto x^{-\alpha}$

- ▶ α : the tail index; a small α means a heavy tail; $\mathbb{E}|X|^p$ is finite only for $p < \alpha$
- ▶ the observations are independent

A5 [Solved]: The Hill Estimator, Step by Step — Task (2/2)

- ▣ **Hill estimator Hill (1975):** the inverse of the mean log-ratio between the k largest values and the threshold

$$\hat{\alpha} = \left(\frac{1}{k} \sum_{i=1}^k \ln \frac{X_{(i)}}{X_{(k+1)}} \right)^{-1}$$

► a heavy tail has large log-ratios, hence a small $\hat{\alpha}$

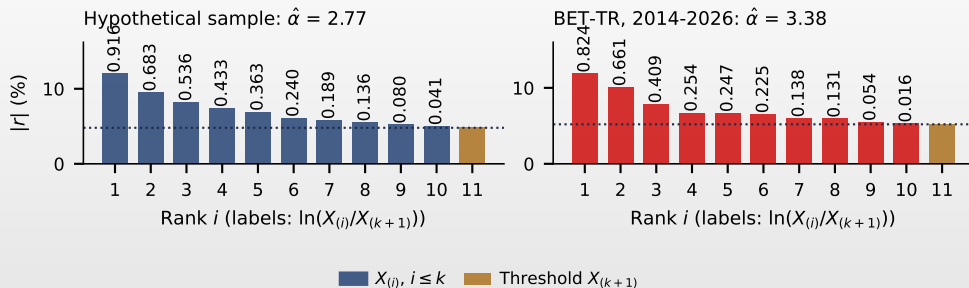
▣ **Tasks**

1. For $k = 10$ compute the log-ratios $\ln(X_{(i)}/X_{(11)})$, their mean and $\hat{\alpha} = 1/\text{mean}$.
2. Compute the standard error $\text{SE} = \hat{\alpha}/\sqrt{k}$ (valid for i.i.d. observations) and the 95% interval $\hat{\alpha} \pm 1.96 \text{ SE}$.
3. State which moments $\mathbb{E}|X|^p$ ($p < \alpha$) exist at the point estimate.
4. State which of these claims hold at 95%.
5. Repeat steps 1–2 on the 11 largest $|r_t|$ of BET-TR, using the worksheet printed by the notebook.

- ▣ **Report:** $\hat{\alpha}$, SE and interval for both samples

- ▣ Notebook: section [A5](#)

A5 [Solved]: Solution (1/2) — Hypothetical Sample



1. The labels are the log-ratios to 4.8: sum 3.616, mean 0.3616, $\hat{\alpha} = 1/0.3616 = 2.77$
2. $SE = 2.77/\sqrt{10} = 0.87$; interval [1.05; 4.48]
 ► a rough approximation by the Normal distribution, valid for i.i.d. Pareto tails
3. At 2.77 the variance exists and the fourth moment does not
4. The interval contains values below 2 and above 4: neither claim holds at 95%

A5 [Solved]: Solution (2/2) — BET-TR Worksheet

i	Date	r_t (%)	$X_{(i)}$ (%)	$\ln(X_{(i)}/X_{(11)})$
1	19 Dec 2018	-11.857	11.857	0.824
2	16 Mar 2020	-10.075	10.075	0.661
3	9 Mar 2020	-7.827	7.827	0.409
4	24 Dec 2018	+6.707	6.707	0.254
5	9 Mar 2022	+6.662	6.662	0.247
6	24 Aug 2015	-6.512	6.512	0.225
7	24 Mar 2020	+5.972	5.972	0.138
8	17 Mar 2020	+5.928	5.928	0.131
9	4 Mar 2022	-5.490	5.490	0.054
10	12 Mar 2020	-5.285	5.285	0.016
11	18 Jan 2016	-5.202	5.202	threshold

- Row 1 worked: $\ln(11.857/5.202) = \ln 2.279 = 0.824$
- Sum of the ten log-ratios 2.958;
 $\hat{\alpha}_{10} = 10/2.958 = 3.38$
- $SE = 3.38/\sqrt{10} = 1.07$; interval [1.28; 5.48]
- Seven of the ten days fall in December 2018 or March 2020
 - ▶ the extremes cluster, so the i.i.d. SE understates the uncertainty
- Interpretation: ten observations cannot decide whether the fourth moment exists
 - ▶ B6 uses $k = 2\%$ of n

B6 [Solved]: Hill on Both Tails, BET-TR and Bitcoin — Task (1/2)

- **Question:** are large losses more frequent than large gains of the same size?
- **Data:** r_{bet} ($n = 2\,992$), r_{btc} ($n = 4\,384$)
 - ▶ losses $\text{loss} = -r[r < 0]$, gains $\text{gain} = r[r > 0]$
 - ▶ each sorted in decreasing order
- **Notation**
 - ▶ $\hat{\alpha}(k)$: the Hill estimator of A5, computed from the k largest values of one tail
 - ▶ $\hat{\alpha}_-$: the tail index of losses; $\hat{\alpha}_+$: that of gains; $\hat{\alpha}_- < \hat{\alpha}_+$ means losses have the heavier tail
 - ▶ $\lfloor \cdot \rfloor$: the integer part
- **Tasks**
 1. Plot the Hill estimate $\hat{\alpha}(k)$ of each tail for k from 1% to 10% of the full-sample n , with the band $\hat{\alpha}(1 \pm 1.96/\sqrt{k})$.
 2. Compute $\hat{\alpha}_-$ and $\hat{\alpha}_+$ at $k = \lfloor 0.02 n \rfloor$ (BET-TR 59, Bitcoin 87), with 95% intervals.
 - the lecture uses $k = 2.5\%$ of n for $|r_t|$, which pools both tails

B6 [Solved]: Hill on Both Tails, BET-TR and Bitcoin — Task (2/2)

▣ **Tasks** (continued)

3. Compute the approximate independent-tail statistic and its two-sided p-value.

$$z = \frac{\hat{\alpha}_- - \hat{\alpha}_+}{\sqrt{\hat{\alpha}_-^2/k + \hat{\alpha}_+^2/k}}$$

- denominator: the standard error of the difference, from $SE = \hat{\alpha}/\sqrt{k}$ for each tail
- under $H_0: \alpha_- = \alpha_+$, $z \approx N(0, 1)$; $|z| > 1.96$: reject H_0 at 5%

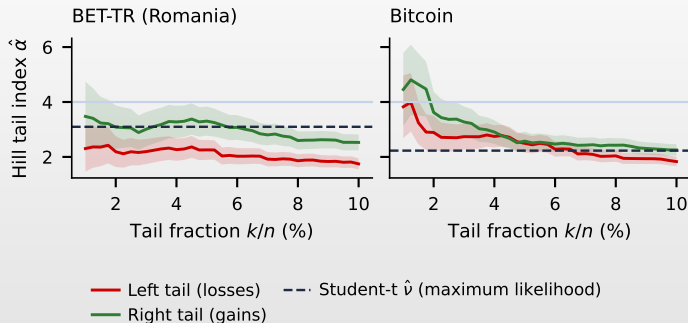
4. Compare with the Student- t $\hat{\nu}$ fitted to the standardised returns: 3.10 for BET-TR (B2), 2.23 for Bitcoin.

▣ **Interpretation**

- ▶ is the gain/loss asymmetry statistically established?
- ▶ does clustering make the true SE larger or smaller?

▣ Notebook: section B6

B6 [Solved]: Solution (1/2) — Hill Plots and Estimates



- BET-TR: $\hat{\alpha}_- = 2.19$ [1.63; 2.75], $\hat{\alpha}_+ = 3.08$ [2.30; 3.87]
- Bitcoin: $\hat{\alpha}_- = 2.89$ [2.28; 3.50], $\hat{\alpha}_+ = 3.62$ [2.86; 4.38]
- E.g. BET-TR losses: $2.19 \times (1 \pm 1.96/\sqrt{59}) = 2.19 \times (1 \pm 0.255)$

B6 [Solved]: Solution (2/2) — Test and Interpretation

- ▣ BET-TR: $\sqrt{2.19^2/59 + 3.08^2/59} = \sqrt{0.0812 + 0.1612} = 0.492$; $z = (2.19 - 3.08)/0.492 = -1.82$, $p = 0.069$
- ▣ Bitcoin: $\sqrt{2.89^2/87 + 3.62^2/87} = 0.497$; $z = (2.89 - 3.62)/0.497 = -1.47$, $p = 0.14$
- ▣ Decision: losses are heavier in both markets, but $H_0: \alpha_- = \alpha_+$ is not rejected at 5%
 - ▶ the statistic treats the two tails as independent samples
 - ▶ extremes cluster in time, so the true SE is larger and the evidence weaker
 - ▶ a joint block bootstrap of the original returns keeps both features (extension)
- ▣ $\hat{\nu}$ is fitted to the whole distribution, while $\hat{\alpha}$ uses only the tail
 - ▶ Bitcoin's 2.23 lies below both tail indices
 - ▶ the Student- t does not fit centre and tails together

A6 [Proposed]: Widening of the Autocorrelation Band under Clustering — Task (1/2)

- **Model:** A4 [Solved]; notebook section A6
- **Setting:** $x_t = \sigma_t \varepsilon_t$ with ε_t i.i.d. $N(0, 1)$, independent of the stationary volatility process
 - ▶ then x_t is uncorrelated
 - ▶ $\text{Var}(\hat{\rho}_1) \approx v_1/T$, with $v_1 = \mathbb{E}[x_t^2 x_{t-1}^2] / (\mathbb{E}x_t^2)^2$ Lobato, Nankervis & Savin (2001)
 - ▶ v_1 : the factor by which volatility clustering inflates the variance of $\hat{\rho}_1$ relative to the i.i.d. value $1/T$
- **Inputs**
 - ▶ A4: $\mathbb{E}\sigma^2 = 1.6$, $\text{Var}(\sigma^2) = 4.0 - 1.6^2 = 1.44$ (in %⁴)
 - ▶ persistent regimes (A9): $\text{Cov}(\sigma_t^2, \sigma_{t-1}^2) = \lambda \text{Var}(\sigma^2)$ with $\lambda = 0.875$
 - ▶ $\lambda \in [0, 1)$: the persistence of the regimes; $\lambda = 0$: states independent across days
 - ▶ S&P 500: $T = 9\,245$, $\hat{\rho}_1 = -0.079$, $T\hat{\tau}_1 = 5.00$

A6 [Proposed]: Widening of the Autocorrelation Band under Clustering — Task (2/2)

▣ Tasks

1. Show that $v_1 = 1$ for the A4 mixture, whose states are independent across days.
2. Show that $v_1 = 1 + \lambda \text{Var}(\sigma^2)/(\mathbb{E}\sigma^2)^2$ with persistent regimes.
3. Compute v_1 and the band-widening factor $\sqrt{v_1}$.
4. Compute the S&P 500 lag-1 statistic $t = \hat{\rho}_1/\text{SE}$ with the i.i.d. SE $1/\sqrt{T}$ and with the robust SE $\sqrt{\hat{\tau}_1}$ ($|t| > 1.96$: significant at 5%).
5. Compute the two 95% band half-widths.

▣ **Interpretation:** which feature widens the band: heavy tails or persistent volatility?

After the Lecture: Remaining Tasks

- These three tasks use tools that [Chapter 1](#) introduces
 - ▶ [A5](#) [Solved] and [B6](#) [Solved]: the Hill estimator of the tail index, step by step and on both tails
 - ▶ [A6](#) [Proposed]: volatility clustering, not heavy tails, widens the autocorrelation band of [B4](#)
- Work on them after the lecture, before the next seminar
- The solution of [A6](#) is discussed at the next seminar

Extension Packs: Content and Prerequisites

Pack	Exercises	Prerequisites
1. Return arithmetic and drawdowns	A1(c) [Solved], A2(b–e), A7	A1, A2(a)
2. Mixtures, regimes and tails	A8 [Solved], A9, A10, A11	A4, A5, A6
3. Robust moments and model checks	B1(d) [Solved], B7, B8	B1, B4, B6
4. Dependence and asymmetry	B9, B10, B11, B12	B3, B4
5. Performance and range estimators	B13, B14, B15	A1, A8, B1
6. Projects	C2, C3	B4, the lecture case study

- Notebook sections carry the same labels as the exercises
- Solutions of [Proposed] exercises are discussed at the next seminar

A1(c) [Solved]: Portfolio Returns — Task and Solution

▣ **Question:** which return adds up across assets?

▣ **Inputs:** 60% in asset A and 40% in asset B for one day; $R_A = +5\%$, $R_B = -2\%$

▣ **Tasks**

1. Compute the portfolio simple return $R_p = 0.6R_A + 0.4R_B$.
2. Compute the weighted log return $0.6r_A + 0.4r_B$, with $r_A = \ln(1 + R_A)$, $r_B = \ln(1 + R_B)$.
3. Compute the true portfolio log return $\ln(1 + R_p)$.

▣ **Solution**

1. $R_p = 0.6 \times 5\% + 0.4 \times (-2\%) = 2.20\%$
2. $0.6 \times \ln 1.05 + 0.4 \times \ln 0.98 = 0.6 \times 4.879\% - 0.4 \times 2.020\% = 2.119\%$
3. $\ln 1.022 = 2.176\%$

▣ Interpretation: simple returns add across assets, log returns add over time (A1)

- ▶ for daily moves the gap between the two rules is small
- ▶ for large moves it is not

A2(b–e) [Proposed]: A Benchmark for Drawdowns — Task (1/2)

- ▣ **Model:** A2(a); notebook section A2
- ▣ **Assumption:** the log price is $X_t = \mu t + \sigma W_t$
 - ▶ W_t a standard Brownian motion: independent increments, $W_t \sim N(0, t)$
 - ▶ μ : the annual drift of the log price; σ : the annual volatility; T : the horizon, in years
 - ▶ maximum log drawdown $D = \max_{s \leq t \leq T} (X_s - X_t) = -\ln(1 + MDD)$: the largest fall of X from a peak at s to a later time t
 - ▶ for $\mu = 0$: $\mathbb{E}[D] = \sqrt{\pi/2} \sigma \sqrt{T}$ Magdon-Ismail et al. (2004), with $\sqrt{\pi/2} = 1.2533$
- ▣ **Inputs:** gold spot, 1990–2026
 - ▶ $\sigma = 16.3\%$, $T = 36.7$ years, mean log return 6.5% a year
 - ▶ observed $MDD = -44.7\%$, so $D = 0.592$

A2(b–e) [Proposed]: A Benchmark for Drawdowns — Task (2/2)

▣ Tasks

1. Use $W(cT) \stackrel{d}{=} \sqrt{c} W(T)$ to explain why $\mathbb{E}[D]$ is proportional to $\sqrt{T}$ when $\mu = 0$ ($c > 0$: the time-scaling factor; $\stackrel{d}{=}$: same distribution).
2. Compute the ratio of $\mathbb{E}[D]$ for 40 and 10 years of the same process.
3. Compute the driftless benchmark for gold and compare it with the observed D .
4. Simulate 2000 daily paths (seed 42) with $\mu = 0$ and with $\mu = 6.5\%$.
5. Plot the mean D against T for both drifts.

- ▣ **Interpretation:** what do steps 1–5 imply for Calmar ratios computed on samples of different length?

A7 [Proposed]: The $\sqrt{h}$ Rule under Autocorrelation — Task (1/2)

- ▣ **Model:** A1 [Solved]; notebook section A7
- ▣ **Assumption:** returns follow a stationary AR(1) process

$$r_t = \rho r_{t-1} + e_t, \quad |\rho| < 1$$

- ▶ e_t : a shock uncorrelated over time, with mean 0
- ▶ ρ : the autoregressive coefficient, equal to the lag-1 autocorrelation; $\rho > 0$: returns of the same sign tend to follow each other
- ▶ $\text{Var}(r_t) = \sigma^2$; the autocovariance at lag j is $\gamma_j = \text{Cov}(r_t, r_{t-j}) = \sigma^2 \rho^j$, and $\gamma_0 = \sigma^2$
- ▣ **The $\sqrt{h}$ rule:** the variance of an h -day return is $h\sigma^2$, true only for uncorrelated returns
- ▣ **Variance ratio:** $VR(h) = \text{Var}(\sum_{j=1}^h r_{t+j}) / (h\sigma^2)$
 - ▶ $VR(h) = 1$ for uncorrelated returns; $VR(h) > 1$ for $\rho > 0$, so the $\sqrt{h}$ rule understates risk
- ▣ **Scaffold, $h = 3$:** $\text{Var}(r_1 + r_2 + r_3) = 3\gamma_0 + 2(2\gamma_1 + \gamma_2)$: two pairs at lag 1, one pair at lag 2

A7 [Proposed]: The $\sqrt{h}$ Rule under Autocorrelation — Task (2/2)

▣ Tasks

1. Generalise the pair count: show that $\text{Var}(\sum_{j=1}^h r_{t+j}) = h\sigma^2[1 + 2\sum_{j=1}^{h-1}(1 - j/h)\rho^j]$.
2. Using $\sum_{j=1}^{h-1} \rho^j = \rho(1 - \rho^{h-1})/(1 - \rho)$ and $\sum_{j \geq 1} j|\rho|^j = |\rho|/(1 - |\rho|)^2$, show that $VR(h) \rightarrow (1 + \rho)/(1 - \rho)$ as $h \rightarrow \infty$.
3. Illustrative calculation: compute $\sqrt{VR(252)}$ and the corrected annual volatilities, treating the sample lag-1 autocorrelations as AR(1) parameters.
 - EUR/RON: $\rho = 0.170$, volatility 4.44%
 - S&P 500: $\rho = -0.079$, volatility 17.99%
4. Explain why volatility clustering alone does not change step 1.

▣ **Interpretation:** which ACF check must precede step 3?

A8 [Solved]: Parkinson Volatility and Its Efficiency — Task (1/2)

- ▣ **Question:** how much information does the daily range carry about volatility?
- ▣ **Notation**
 - ▶ O_t, H_t, L_t, C_t : the open, high, low and close of day t
 - ▶ their logs: $o_t = \ln O_t, h_t = \ln H_t, l_t = \ln L_t, c_t = \ln C_t$
 - ▶ the log range: $h_t - l_t = \ln(H_t/L_t)$
- ▣ **Inputs:** H/L of five days: 1.020, 1.015, 1.030, 1.010, 1.025; $A = 252$
- ▣ **Assumption:** within the day the log price is a driftless Brownian motion with volatility σ
 - ▶ no overnight jumps
- ▣ **Parkinson estimator** [Parkinson \(1980\)](#): the daily variance estimated from the range

$$\hat{\sigma}_P^2 = \frac{\overline{(h - l)^2}}{4 \ln 2}$$

- ▶ $\overline{(h - l)^2}$: the average over days of the squared range
- ▶ $4 \ln 2 = 2.7726$: under the assumption $\mathbb{E}[(h - l)^2] = 4 \ln 2 \sigma^2$, so the division makes the estimator unbiased

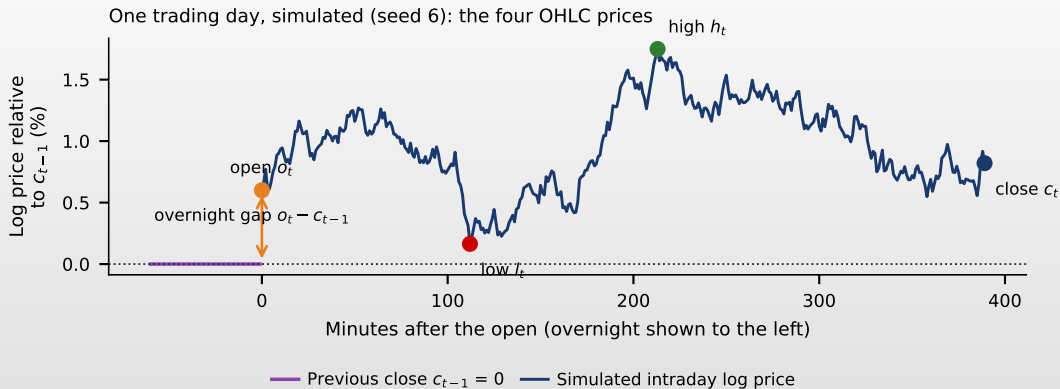
A8 [Solved]: Parkinson Volatility and Its Efficiency — Task (2/2)

▣ Tasks

1. Complete a five-row table: $\ln(H/L)$, its square, and the mean of the squares.
2. Compute $\hat{\sigma}_P^2$, the daily and the annual volatility.
3. Name the information the estimator ignores.
4. Explain the effect of a strong drift on the estimate.
5. Using $\mathbb{E}[(h-l)^2] = 4 \ln 2 \sigma^2$ and $\mathbb{E}[(h-l)^4] = 9\zeta(3) \sigma^4$, compare the per-day variance of $(h-l)^2/(4 \ln 2)$ with $\text{Var}(r_t^2) = 2\sigma^4$.
 - $\zeta(3) = \sum_{n \geq 1} n^{-3} = 1.202$: the Riemann zeta function at 3
 - r_t^2 : the close-to-close estimator of the daily variance, from a return with distribution $N(0, \sigma^2)$
 - the estimator with the smaller variance is more efficient: it needs fewer days for the same precision

▣ Notebook: section [A8](#)

A8 [Solved]: Solution (1/3) — The Estimate



A8 [Solved]: Solution (2/3) — The Estimate

H/L	$\ln(H/L)$	square ($\times 10^{-4}$)
1.020	0.0198	3.92
1.015	0.0149	2.22
1.030	0.0296	8.74
1.010	0.0100	0.99
1.025	0.0247	6.10
mean		4.39

- A simulated day: the range $h_t - l_t$ uses the whole path, the close-to-close return only its end points

2. $\hat{\sigma}_P^2 = 0.000439/2.7726 = 0.000158$; $\hat{\sigma}_P = 1.26\%$ a day, $1.26\% \times 15.87 = 20.0\%$ a year
3. Ignored: the open and the overnight jump Yang & Zhang (2000)
4. A drift widens the range and biases $\hat{\sigma}_P$ up Rogers & Satchell (1991)

A8 [Solved]: Solution (3/3) — Efficiency

5. $\text{Var}((h - l)^2) = (9\zeta(3) - 16(\ln 2)^2)\sigma^4 = (10.819 - 7.687)\sigma^4 = 3.131\sigma^4$

▶ divided by $(4\ln 2)^2 = 7.687$: $0.407\sigma^4$ per day for the Parkinson term

▶ close-to-close: $\text{Var}(r_t^2) = 3\sigma^4 - \sigma^4 = 2\sigma^4$ per day

□ For an n -day average of independent days, both variances are divided by n

▶ the ratio stays $2/0.407 = 4.9$

□ One day of ranges carries about the information of five days of close-to-close returns

□ Interpretation: the gain assumes driftless continuous trading without overnight jumps

▶ a stale open (an open computed from prices of components that have not yet traded) affects the estimators that use the open

▶ namely Garman–Klass, Rogers–Satchell and Yang–Zhang, not Parkinson (B15)

A9 [Proposed]: The Origin of the Autocorrelation of r_t^2 — Task (1/2)

- ▣ **Model:** A4 [Solved] and A6; notebook section A9
- ▣ **Assumption:** as in A4, $\sigma_t^2 \in \{1, 4\}$ (in %²) with stationary probabilities (0.8; 0.2)
 - ▶ but the state follows a two-state Markov chain [Hamilton \(1989\)](#): tomorrow's state depends only on today's
 - ▶ transition matrix $\begin{pmatrix} 0.975 & 0.025 \\ 0.1 & 0.9 \end{pmatrix}$ (rows: today calm L , turbulent H)
 - ▶ $p_{LL} = 0.975$ and $p_{HH} = 0.9$ are the probabilities of staying in the same state
 - ▶ $\pi_L = 0.8$, $\pi_H = 0.2$: the stationary probabilities, i.e. the long-run shares of the two states
- ▣ **Tasks**
 1. Check the balance condition $\pi_L(1 - p_{LL}) = \pi_H(1 - p_{HH})$, i.e. $0.8 \times 0.025 = 0.2 \times 0.1$: the flow from L to H equals the flow from H to L .
 2. Compute $\lambda = p_{LL} + p_{HH} - 1$, the persistence of the chain: $\lambda = 0$ for independent states, $\lambda \rightarrow 1$ for very persistent regimes.

A9 [Proposed]: The Origin of the Autocorrelation of r_t^2 — Task (2/2)

▣ **Tasks** (continued)

3. With the indicator $I_t = 1$ in the turbulent state, $\sigma_t^2 = 1 + 3I_t$; show that $\text{Cov}(I_t, I_{t-k}) = \pi_H(1 - \pi_H)\lambda^k$.
4. Deduce that $\text{Cov}(r_t^2, r_{t-k}^2) = \text{Var}(\sigma^2)\lambda^k$ for $k \geq 1$.
5. Using A4, compute $\text{Var}(\sigma^2)$, $\text{Var}(r^2) = 3\mathbb{E}\sigma^4 - (\mathbb{E}\sigma^2)^2$ and $\text{corr}(r_t^2, r_{t-k}^2)$ at lags 1 and 10.
6. Plot the implied ACF against the S&P 500 ACF of r_t^2 (0.31 at lag 1, 0.04 at lag 100).

▣ **Interpretation:** which two features of the data can this model not reproduce?

A10 [Proposed]: The S&P 500 Tails in a GARCH with Normal Innovations — Task (1/2)

▣ **Model:** A4 [Solved]; notebook section A10

▶ an advanced preview of Chapter 5

▣ **Model equations** GARCH(1,1) [Bollerslev \(1986\)](#): today's variance depends on yesterday's shock and yesterday's variance

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

- ▶ μ : the mean return; ε_t : the shock of day t ; z_t : the standardised innovation, i.i.d. $N(0, 1)$
- ▶ σ_t^2 : the conditional variance, known at the end of day $t - 1$
- ▶ $\omega > 0$: the base level of the variance; $\alpha \geq 0$: the reaction to yesterday's shock; $\beta \geq 0$: the persistence of yesterday's variance
- ▶ $\alpha + \beta < 1$: the variance reverts to its long-run mean $\omega/(1 - \alpha - \beta)$
- ▶ here α is a GARCH coefficient, not a tail index

A10 [Proposed]: The S&P 500 Tails in a GARCH with Normal Innovations — Task (2/2)

- ▣ **Result:** if $1 - (\alpha + \beta)^2 - 2\alpha^2 > 0$, the unconditional kurtosis of r_t is

$$K = \frac{3[1 - (\alpha + \beta)^2]}{1 - (\alpha + \beta)^2 - 2\alpha^2} > 3$$

▣ **Tasks**

1. Compute K for $(\alpha, \beta) = (0.10; 0.85)$ and $(0.10; 0.88)$.
2. State the condition for a finite fourth moment.
3. Compare it with $\alpha + \beta < 1$.
4. With $\alpha + \beta = 0.99$, solve for the α that gives the S&P 500 kurtosis $K = 13.9$.
5. Fit the model to `r_sp` in %.
 - `arch_model(100*r_sp, mean='Constant', vol='GARCH', p=1, q=1, dist='normal')`
6. Compute the implied K .
7. Draw a QQ plot of $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$.

- ▣ **Interpretation:** what does the QQ plot say about the Normal distribution of z_t ?

A11 [Proposed]: The Hill Estimator as a Maximum Likelihood Estimator — Task (1/2)

- ▣ **Model:** A5 [Solved]; notebook section A11
- ▣ **Assumption:** given the threshold $u = X_{(k+1)}$, the k exceedances are the order statistics of an i.i.d. Pareto sample
 - ▶ Pareto density $f(x) = \alpha u^\alpha x^{-\alpha-1}$, $x > u$, with tail index α
- ▣ **Concepts**
 - ▶ the log-likelihood $\ell(\alpha) = \sum_{i=1}^k \ln f(X_{(i)})$: how plausible the data are for a given value of α
 - ▶ the maximum likelihood estimator: the value of α that maximises $\ell(\alpha)$
 - ▶ ℓ' , ℓ'' : the first and second derivatives with respect to α
 - ▶ the Fisher information $-\mathbb{E} \ell''(\alpha)$; $\text{SE}(\hat{\alpha}) \approx 1/\sqrt{\text{information}}$

A11 [Proposed]: The Hill Estimator as a Maximum Likelihood Estimator — Task (2/2)

▣ Tasks

1. Write the log-likelihood $\ell(\alpha)$ of the k exceedances (the ordering adds only the constant $\ln k!$).
2. Solve $\ell'(\alpha) = 0$ and show that the maximiser is the Hill estimator $\hat{\alpha} = k / \sum_{i=1}^k \ln(X_{(i)}/u)$ Hill (1975).
3. Compute $\ell''(\alpha)$, the information k/α^2 and $\text{SE}(\hat{\alpha}) \approx \hat{\alpha}/\sqrt{k}$.
4. Check the SE on the data of A5.
5. Simulate $n = 2\,000\,000$ values $|X|$, X Student- t with $\nu = 3$ (`numpy.random.default_rng(1)`).
6. Plot $\hat{\alpha}_k$ (the Hill estimate from k values) against k/n and report $k = 200, 2\,000, 20\,000, 200\,000$.

▣ **Interpretation:** why does $\hat{\alpha}_k$ drift away from 3 when k is large?

B1(d) [Solved]: Quantile-Based Moments with Bootstrap Intervals (1/2)

- ▣ **Question:** do negative skewness and excess kurtosis survive measures that need no moments?
- ▣ **Measures** Kim & White (2004), both 0 for the Normal distribution (q_p : the quantile of order p)
 - ▶ Hinkley skewness $\frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$: the position of the median between $q_{0.05}$ and $q_{0.95}$
 - ▶ it lies in $[-1, 1]$; negative: the left tail is longer than the right one
 - ▶ Crow–Siddiqui kurtosis $\frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$ (Prerequisites for Today, 2/10)
- ▣ **Block bootstrap:** 1 000 resamples (`default_rng(42)`)
 - ▶ draw $\lceil T/20 \rceil$ starting days uniformly ($\lceil \cdot \rceil$: rounding up) and take 20 consecutive returns from each
 - ▶ the blocks keep the volatility clustering inside them
 - ▶ cut to T and recompute both measures
 - ▶ the 95% percentile interval: the 2.5% and 97.5% quantiles of the 1 000 recomputed values

B1(d) [Solved]: Quantile-Based Moments with Bootstrap Intervals (2/2)

	Moment skew.	Hinkley [95% CI]	Excess kurt.	Crow–Siddiqui [95% CI]
BET-TR	−1.40	−0.036 [−0.085, 0.025]	17.8	1.06 [0.73, 1.43]
Bitcoin	−0.70	−0.018 [−0.069, 0.032]	12.0	2.25 [1.88, 2.64]
EUR/RON	0.73	0.041 [−0.022, 0.093]	15.1	4.87 [3.74, 6.13]

- Every kurtosis interval lies above 0: heavy tails are established without moments
- Moment skewness and Hinkley skewness measure different things
 - ▶ every Hinkley interval contains 0
 - ▶ this does not establish symmetry

B7 [Proposed]: A Kurtosis That Does Not Converge — Task

- ▣ **Model:** B1 [Solved]; notebook section B7
- ▣ **Data:** r_{sp} ; window n = the returns dated before 3 January of year 1990 + n , $n = 1, \dots, 36$
 - ▶ 253 returns for $n = 1$, 2 527 for $n = 10$, 9 067 for $n = 36$
- ▣ **Simulation:** 500 Student- t paths of length 9 245 for $\nu = 3$ and for $\nu = 6$ (`default_rng(42)`)
 - ▶ cut at the same window counts
- ▣ **Tasks**
 1. Compute the excess kurtosis of the S&P 500 on each window.
 2. List the three largest jumps, with the days that cause them.
 3. For each ν compute the 5%, 50% and 95% quantiles of the simulated kurtosis at each n .
 4. Plot the data curve over both bands on a symmetric log scale.
- ▣ **Interpretation**
 - ▶ why does the $\nu = 6$ band settle slowly although the kurtosis of t_6 is finite?
 - ▶ what does “excess kurtosis = 10.9” then mean?

B8 [Proposed]: Stylised Facts Reproduced by GARCH- t — Task (1/2)

▣ **Model:** B3 and B6 [Solved], A10; notebook section B8

▣ **Data:** r_{sp} converted to % ($T = 9\,245$)

- ▶ constant mean
- ▶ ω in %² a day

```
from arch import arch_model
from arch.univariate import StudentsT
am = arch_model(100 * r_sp, mean='Constant', vol='GARCH', p=1, q=1, dist='t')
res = am.fit(dis='off')
am.distribution = StudentsT(seed=np.random.default_rng(42)) # reproducible paths
sim = am.simulate(res.params, nobs=len(r_sp), burn=1000)['data']
```

B8 [Proposed]: Stylised Facts Reproduced by GARCH- t — Task (2/2)

□ Tasks

1. Fit the model and list all five parameters with units.
2. Pilot: simulate one path and compute its excess kurtosis.
3. Then simulate 200 paths.
4. For each path compute these statistics.
 - the excess kurtosis
 - the ACF of $|r_t|$ at lags 1, 20, 100, 250
 - the Hill $\hat{\alpha}$ of $|r_t|$ at $k = 2\%$ of T
 - $L(1) = \text{corr}(r_t, |r_{t+1}|)$: the correlation between today's return and tomorrow's size (the leverage effect, B9)
5. Build the pointwise 5–95% intervals.
6. Mark each observed value as inside or outside.

□ Interpretation

- ▶ which facts are not reproduced?
- ▶ which model component would you change for each?

B9 [Proposed]: Leverage Effect, BET-TR versus Bitcoin — Task (1/2)

- ▣ **Model:** B3 [Solved]; notebook section B9
- ▣ **Lag convention:** $L(k) = \text{corr}(r_t, |r_{t+k}|)$ Christie (1982)
 - ▶ $L(k) < 0$: after falls, later returns are larger in absolute value (the leverage effect)
 - ▶ k counts observation days of each series
 - ▶ $k > 0$: today's return and a later size
 - ▶ example, $k = 1$: the BET-TR return of Monday 16 Mar 2020 is paired with $|r|$ of Tuesday 17 Mar 2020
- ▣ **Data:** `r_bet` and `r_btc`, each on its own calendar
- ▣ **Tasks**
 1. Compute $L(k)$ for $k = -20, \dots, 20$.
 2. Plot both series with the i.i.d. reference bands $\pm 1.96/\sqrt{T}$ (0.036 and 0.030).

B9 [Proposed]: Leverage Effect, BET-TR versus Bitcoin — Task (2/2)

▣ **Tasks** (continued)

3. Report $L(1)$ and $L(5)$ for both markets.
4. State whether they lie outside the reference bands.
5. Give two competing explanations for a weaker Bitcoin effect.

▣ **Interpretation**

- ▶ why is a reference band not a confidence interval under clustering?
- ▶ what does B10 add?

B10 [Proposed]: The Leverage Effect as a Regression — Task (1/2)

- ▣ **Model:** B4 [Solved], B9; notebook section B10
- ▣ **Regression:** $|r_{t+1}| = c + \beta^+ r_t^+ + \beta^- r_t^- + \sum_{j=1}^5 \delta_j |r_{t-j}| + u_{t+1}$
 - ▶ $r_t^+ = \max(r_t, 0) \geq 0$, $r_t^- = \min(r_t, 0) \leq 0$: the positive and the negative part of the return
 - ▶ c : the constant; δ_j : the effect of $|r_{t-j}|$, controls for volatility clustering; u_{t+1} : the error
 - ▶ a gain $+a$ adds $\beta^+ a$, a loss $-a$ adds $-\beta^- a$
 - ▶ a symmetric response means $\beta^+ = -\beta^-$
 - ▶ design row for day t : $(1, r_t^+, r_t^-, |r_{t-1}|, \dots, |r_{t-5}|)$
 - ▶ response: $|r_{t+1}|$
- ▣ **Data:** r_sp, r_bet, r_btc, r_eur

B10 [Proposed]: The Leverage Effect as a Regression — Task (2/2)

▣ Tasks

1. Estimate by OLS with Newey–West standard errors [Newey & West \(1987\)](#), robust to heteroskedasticity and autocorrelation, with lag truncation $\lfloor 4(T/100)^{2/9} \rfloor$ (the number of error lags taken into account).
2. Test $H_0: \beta^+ + \beta^- = 0$ (a symmetric response) at 5%, against χ_1^2 , with the Wald statistic
 - $W = (\hat{\beta}^+ + \hat{\beta}^-)^2 / [\text{Var}(\hat{\beta}^+) + \text{Var}(\hat{\beta}^-) + 2 \text{Cov}(\hat{\beta}^+, \hat{\beta}^-)]$
 - denominator: the variance of $\hat{\beta}^+ + \hat{\beta}^-$, from the Newey–West covariance matrix; W is the square of a t -statistic
3. Plot the predicted $|r_{t+1}|$ after a gain $+a$ and a loss $-a$, lag controls at their means.
4. Show why $|r_t|$ cannot be added as a control.

B11 [Proposed]: EUR/RON Across Regimes — Task (1/3)

▣ **Model:** B1, B4 [Solved]; notebook section B11

▣ **Data:** `r_eur` (BNR reference rate), split into three periods

- ▶ Jul 2005 – 2011, 2012–2019, 2020–2026: 1 656, 2 013, 1 683 returns
- ▶ lags built inside each period

▣ **Tasks**

1. In each period regress $r_t = a + \rho r_{t-1} + e_t$ by OLS, with a HAC SE (lag truncation $\lfloor 4(T/100)^{2/9} \rfloor$).
 - a : the constant; ρ : the lag-1 autocorrelation; e_t : the error
 - HAC SE: a standard error robust to heteroskedasticity and autocorrelation (as in B10)
2. Report $\hat{\rho}$, the SE, the volatility and the excess kurtosis of each period.

B11 [Proposed]: EUR/RON Across Regimes — Task (2/3)

▣ **Tasks** (continued)

3. Test equal ρ across periods with the statistic below.

$$W = \sum_j w_j (\hat{\rho}_j - \bar{\rho})^2 \sim \chi_2^2$$

- $j = 1, 2, 3$: the period; weights $w_j = 1/\text{SE}_j^2$, so precise estimates count more
- $\bar{\rho} = \sum_j w_j \hat{\rho}_j / \sum_j w_j$: the weighted mean of the three estimates
- W measures the spread of the estimates around $\bar{\rho}$; 2 degrees of freedom = 3 periods — 1

B11 [Proposed]: EUR/RON Across Regimes — Task (3/3)

▣ **Tasks** (continued)

4. Separately, locate one variance break with the cumulative sum of squares (CUSUM) [Inclán & Tiao \(1994\)](#).

- $C_k = \sum_{t \leq k} x_t^2$: the sum of squares up to day k
- $D_k = C_k/C_T - k/T$: the share of total variance accumulated by day k , minus the share of time elapsed
- $D_k \approx 0$ for every k if the variance is constant
- the break is at $\arg \max_k |D_k|$, the day k where $|D_k|$ is largest

▣ **Interpretation**

- ▶ why is the 2020–2026 SE so large?
- ▶ why is the CUSUM date descriptive rather than a test?

B12 [Proposed]: Long Memory or Breaks? — Task (1/3)

- ▣ **Model:** B4, B6 [Solved]; notebook section B12 (helper `local_whittle(x, m)`)
- ▣ **Memory parameter d :** how fast the ACF of $x_t = |r_t|$ decays
 - ▶ $d = 0$: short memory, the ACF decays geometrically
 - ▶ $0 < d < 0.5$: stationary long memory, the ACF decays slowly, like k^{2d-1}
 - ▶ $d \geq 0.5$: the series is no longer stationary
- ▣ **Data:** $|r_t|$ of the S&P 500, BET-TR and Bitcoin

B12 [Proposed]: Long Memory or Breaks? — Task (2/3)

- ▣ **Local Whittle method** [Robinson \(1995\)](#): demean $x_t = |r_t|$
 - ▶ Fourier frequencies $\lambda_j = 2\pi j/T$, $j = 1, \dots, m$; low frequencies correspond to long cycles
 - ▶ periodogram $I(\lambda) = \frac{1}{2\pi T} \left| \sum_t x_t e^{-i\lambda t} \right|^2$: the strength of cycles of frequency λ in the series
 - ▶ long memory makes $I(\lambda_j)$ grow like λ_j^{-2d} at low frequencies
- ▣ $\hat{d}$ minimises over $d \in [-0.49, 0.99]$

$$\ln \left(\frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^m \ln \lambda_j$$

- ▶ $m = \lfloor T^p \rfloor$: the number of low frequencies used; $p \in (0, 1)$ sets the bandwidth
- ▶ $SE = 1/(2\sqrt{m})$

B12 [Proposed]: Long Memory or Breaks? — Task (3/3)

▣ Tasks

1. Estimate d with $p = 0.65$ and give the 95% interval.
2. Plot $\hat{d}$ against p from 0.45 to 0.80, with the lines $d = 0$ and $d = 0.5$.
3. Locate one variance break of r_t (B11).
4. Re-estimate $\hat{d}$ on each side of the break.
5. Plot the ACF of $|r_t|$ before and after the break.

▣ Interpretation

- ▶ is $\hat{d} > 0$ robust to p and to the break?
- ▶ or is it consistent with regimes Diebold & Inoue (2001)?
- ▶ what does $\hat{d} \geq 0.5$ imply?

B13 [Proposed]: BET-TR vs S&P 500, Same Period — Task (1/2)

- ▣ **Model:** A1, B1 [Solved]; notebook section B13
- ▣ **Data:** BET-TR (total return, RON) and S&P 500 (price index, USD)
 - ▶ align the *prices* on common dates (23 Sep 2014 – 18 Sep 2026)
 - ▶ then compute 2 903 returns
 - ▶ interval lengths: 2 231 one-day, 538 of two or three days, 134 longer (at most 6 days)
- ▣ **Tasks**
 1. Plot the rebased wealth (log scale) and the two underwater curves, DD_t of A2(a).

B13 [Proposed]: BET-TR vs S&P 500, Same Period — Task (2/2)

▣ **Tasks** (continued)

2. Compute these measures for both indices.

- $\text{CAGR} = (P_{\text{end}}/P_{\text{start}})^{1/Y} - 1$, with Y the number of calendar years; the volatility of simple returns ($A = 252$)
- the Sharpe ratio $252\bar{R}/(\sqrt{252}\hat{\sigma}_R)$ with $r_f = 0$: the annual return per unit of volatility
- $\bar{R}$, $\hat{\sigma}_R$: the mean and the standard deviation of daily simple returns; r_f : the risk-free rate
- the Sortino ratio [Sortino & van der Meer \(1991\)](#) $252\bar{R}/(\sqrt{252}\hat{\sigma}_D)$, with the downside deviation

$$\hat{\sigma}_D = \sqrt{\frac{1}{T} \sum_t \min(R_t, 0)^2}$$
 over all days
- the Sortino ratio penalises loss days only
- the MDD and the Calmar ratio ([A2\(a\)](#))

▣ **Report**: the chart and a two-column table

▣ **Interpretation**: list three reasons why the comparison is descriptive only (dividends, currency, risk-free rate in RON)

B14 [Proposed]: Significance of the Sharpe Difference — Task (1/2)

▣ **Model:** B1(d), B13; notebook section B14

▣ **Data:** the 2 903 paired returns of B13

- ▶ simple returns for Sharpe ratios (1.36 and 0.75)
- ▶ log returns for correlations

▣ **Tasks**

1. Convert to daily Sharpe ratios ($\widehat{SR}/\sqrt{252}$).
2. Compute the Jobson–Korkie test [Jobson & Korkie \(1981\)](#) with the correction of [Mommel \(2003\)](#).

- $z = (\widehat{SR}_1 - \widehat{SR}_2)/\sqrt{\widehat{V}}$
- $\widehat{V} = \frac{1}{T} [2(1 - \hat{\rho}) + \frac{1}{2}(\widehat{SR}_1^2 + \widehat{SR}_2^2 - 2\widehat{SR}_1\widehat{SR}_2\hat{\rho}^2)]$
- $\widehat{SR}_1, \widehat{SR}_2$: the daily Sharpe ratios of BET-TR and the S&P 500; $\hat{\rho}$: the correlation of their returns
- $\widehat{V}$: the estimated variance of the difference; under H_0 (equal Sharpe ratios) $z \approx N(0, 1)$

B14 [Proposed]: Significance of the Sharpe Difference — Task (2/2)

▣ **Tasks** (continued)

3. Run a paired circular block bootstrap, 2000 resamples (`default_rng(42)`).
 - draw $\lceil T/21 \rceil$ start days, take 21 consecutive days from each (wrapping at the end), cut to T
 - apply the **same** indices to both series
4. Report the 95% percentile interval of the annualised difference.
5. Compute four correlations.
 - same day
 - BET-TR with the S&P 500 of the previous day and of the next day
 - weekly (Friday) returns

▣ **Interpretation**

- ▶ is the difference significant?
- ▶ what do both methods assume?
- ▶ which correlation belongs in a portfolio?

B15 [Proposed]: Range-Based Estimators on the S&P 500 — Task (1/2)

- ▣ **Model:** A8 [Solved]; notebook section B15
 - ▶ tested helpers for all five estimators
 - ▶ formulas in Chapter 1
- ▣ **Data:** S&P 500 index OHLC, 2 Jan 2008 – 18 Sep 2026
- ▣ SPY ETF on one adjusted basis
 - ▶ C^* = the adjusted close (for dividends and splits)
 - ▶ $O^*, H^*, L^* = (C^*/C) \times O, H, L$, with C the unadjusted close and C^*/C the day's adjustment factor
 - ▶ example, 16 Mar 2020: factor $218.08/239.85 = 0.9092$; $O^* = 241.18 \times 0.9092 = 219.29$, $H^* = 233.58$, $L^* = 215.82$

B15 [Proposed]: Range-Based Estimators on the S&P 500 — Task (2/2)

▣ Tasks

1. Compute 21-day rolling close-to-close, Parkinson, Garman–Klass [Garman & Klass \(1980\)](#), Rogers–Satchell [Rogers & Satchell \(1991\)](#) and Yang–Zhang [Yang & Zhang \(2000\)](#) volatilities ($A = 252$).
2. Compute their time averages.
3. For the index and for SPY, compare the 21-day averages of close-to-close and Yang–Zhang with two full-sample volatilities.
 - overnight: $\text{sd}(o_t - c_{t-1})\sqrt{252}$, with $o_t - c_{t-1}$ the overnight log return (notation of [A8](#))
 - open-to-close: $\text{sd}(c_t - o_t)\sqrt{252}$
 - sd: the sample standard deviation

- ▣ **Interpretation:** why does Yang–Zhang stay below close-to-close on the index but not on SPY?

C2 [Proposed]: Did BET-TR Autocorrelation Fall after the FTSE Reclassification? (1/3)

- ▣ **Model:** B4 [Solved], B11; a project seed
- ▣ **Context:** BET-TR has a positive lag-1 autocorrelation, typical of less liquid markets
 - ▶ Romania became a FTSE Russell Secondary Emerging market on 21 Sep 2020
- ▣ **Data:** r_{bet} and a liquidity measure
 - ▶ liquidity: daily traded value (volume $\times$ raw close, in lei) of 11 BVB blue chips
 - ▶ the stocks: TLV, SNP, BRD, TGN, FP, SNG, H2O, SNN, EL, DIGI, TEL
 - ▶ summed over the stocks listed that day
 - ▶ report how many are listed each year

C2 [Proposed]: Did BET-TR Autocorrelation Fall after the FTSE Reclassification? (2/3)

▣ Tasks

1. Compute $\hat{\rho}_1$ on rolling 250-day windows (step 5 days).
2. Add the B4 bands, labelled as reference bands under $\rho_1 = 0$.
3. Estimate ρ_1 before and after 21 Sep 2020 as an AR(1) slope with HAC SEs (B11).
4. Report the statistic for the difference of the two estimates.

$$z = \frac{\hat{\rho}_{\text{after}} - \hat{\rho}_{\text{before}}}{\sqrt{SE_1^2 + SE_2^2}}$$

- SE_1, SE_2 : the HAC standard errors before and after
- under H_0 (no change) $z \approx N(0, 1)$, if the two periods are independent

C2 [Proposed]: Did BET-TR Autocorrelation Fall after the FTSE Reclassification? (3/3)

▣ **Tasks** (continued)

5. Report the interaction e in the regression below (HAC, 10 lags).

$$r_t = a + b r_{t-1} + c \text{post}_t + e r_{t-1} \text{post}_t$$

- $\text{post}_t = 1$ after 21 Sep 2020 and 0 before
- b : the autocorrelation before; e : its change after the reclassification; a , c : the constant and its change

6. Correlate the rolling $\hat{\rho}_1$ with the log 250-day average traded value.

▣ **Deliverable:** a Quantlet folder and one slide

- ▶ a descriptive conclusion and its main limitation
- ▶ not a causal claim

C3 [Proposed]: Is the VIX Mostly Path-Dependent? — Task (1/5)

- **Model:** the lecture case study
 - ▶ a replication project, not an in-class task
- **Question:** can past S&P 500 returns alone explain the VIX, as [Guyon & Lekeufack \(2023\)](#) find?
 - ▶ 94.6% of its variance in 2000–2018, 85.5% in 2019 – May 2022
- **Data:** daily closes of the S&P 500, the VIX and the 9-day VIX
 - ▶ S_{t_i} : the S&P 500 level at the close of day t_i ; simple returns $r_{t_i} = S_{t_i}/S_{t_i-1} - 1$
 - ▶ volatility indices divided by 100
- **Model**, eq. (3.7) in the paper: today's volatility is a linear function of two summaries of past returns

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \Sigma_t$$

- ▶ σ_t : the VIX (or the 9-day VIX) at the close of day t , divided by 100
- ▶ $\beta_0, \beta_1, \beta_2$: coefficients estimated from the data

C3 [Proposed]: Is the VIX Mostly Path-Dependent? — Task (2/5)

- ▣ **Trend feature:** a weighted average of the returns of the last 1000 days

$$R_{1,t} = \sum_{i=0}^{999} K_1(i\Delta t) r_{t-i}$$

- ▶ i : the age of the return, in days; $\Delta t = 1/252$: one day, in years; $K_1(i\Delta t)$: its weight
- ▶ $R_{1,t} < 0$ after a run of falls, $R_{1,t} > 0$ after a run of rises

- ▣ **Volatility feature:** the square root of a weighted average of squared returns

$$\Sigma_t = \left(\sum_{i=0}^{999} K_2(i\Delta t) r_{t-i}^2 \right)^{1/2}$$

- ▶ a weighted historical volatility, with weights K_2

- ▣ **Reading the coefficients**

- ▶ $\beta_1 < 0$: after falls ($R_{1,t} < 0$) volatility rises, i.e. the leverage effect
- ▶ $\beta_2 > 0$: high recent volatility raises today's volatility, i.e. volatility clustering

C3 [Proposed]: Is the VIX Mostly Path-Dependent? — Task (3/5)

- ▣ **Kernels**, eq. (3.9) in the paper: a time-shifted power law

$$K_j(\tau) = \frac{(\tau + \delta_j)^{-\alpha_j}}{Z_j}, \quad j = 1, 2$$

- ▶ τ : the age, in years; Z_j such that $\sum_{i=0}^{999} K_j(i\Delta t)\Delta t = 1$
- ▶ $\alpha_j > 1$: the speed at which the weights decay; a small α_j means long memory
- ▶ $\delta_j > 0$: the shift; a small δ_j puts the weight on the last few days

- ▣ **Parameters and units**

- ▶ β_0 in volatility units ($0.03 = 3$ VIX points)
- ▶ β_1, β_2 unitless
- ▶ δ_j in years

- ▣ **Start** (Table 3 in the paper, VIX row)

- ▶ $\beta_0 = 0.057$, $\beta_1 = -0.095$, $\beta_2 = 0.82$
- ▶ $\alpha_1 = 1.06$, $\delta_1 = 0.020$, $\alpha_2 = 1.60$, $\delta_2 = 0.052$
- ▶ bounds $\alpha_j \in [1.0001, 20]$, $\delta_j \in [10^{-4}, 5]$

C3 [Proposed]: Is the VIX Mostly Path-Dependent? — Task (4/5)

▣ **Tasks**, in stages

1. Fixed kernels: compute $R_{1,t}$ and Σ_t with the Table 3 kernels.
2. Estimate $\beta_0, \beta_1, \beta_2$ by OLS on 2000–2018.
3. Full model: estimate all seven parameters by nonlinear least squares on 1 Jan 2000 – 31 Dec 2018, minimising $\sum_t (\sigma_t - \hat{\sigma}_t)^2$, with $\hat{\sigma}_t$ the model value.
 - use `scipy.optimize.least_squares`
 - recompute the features at every trial value
4. Report R^2 and RMSE on the training set and on 1 Jan 2019 – 15 May 2022.
 - for the full model and for the restrictions $\beta_1 = 0$ and $\beta_2 = 0$
5. Repeat the full model for the 9-day VIX (training from 2011).
6. Keep the parameters fixed and score 16 May 2022 – 18 Sep 2026.

C3 [Proposed]: Is the VIX Mostly Path-Dependent? — Task (5/5)

▣ **Scores**, computed separately on each sample of T days

- ▶ $\text{RMSE} = \sqrt{\frac{1}{T} \sum_t (\sigma_t - \hat{\sigma}_t)^2}$: the typical error, in volatility units
- ▶ $R^2 = 1 - \sum_t (\sigma_t - \hat{\sigma}_t)^2 / \sum_t (\sigma_t - \bar{\sigma})^2$: the share of the variation of σ_t explained by the model
- ▶ $\bar{\sigma}$: the mean of σ_t ; $R^2 = 1$: a perfect fit; $R^2 \leq 0$: no better than the mean

▣ **Report**

- ▶ the parameter table and the scores next to Table 2 of the paper
- ▶ the observed and fitted VIX, and the ratio of observed to fitted VIX
- ▶ the fitted kernels

▣ **Interpretation**: the model fits the VIX on the same day

- ▶ it is not a next-day forecast
- ▶ which feature explains more, $R_{1,t}$ or Σ_t ?

References (1/3)

- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Bollerslev, T., Hood, B., Huss, J., Pedersen, L.H. (2018). Risk everywhere: modeling and managing volatility. *Review of Financial Studies*, 31(7), 2729–2773.
- Christie, A.A. (1982). The stochastic behavior of common stock variances: value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432.
- Clark, P.K. (1973). A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41(1), 135–155.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Dalla, V., Giraitis, L., Phillips, P.C.B. (2022). Robust tests for white noise and cross-correlation. *Econometric Theory*, 38(5), 913–941.
- Diebold, F.X., Inoue, A. (2001). Long memory and regime switching. *Journal of Econometrics*, 105(1), 131–159.
- Dimson, E. (1979). Risk measurement when shares are subject to infrequent trading. *Journal of Financial Economics*, 7(2), 197–226.
- Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction* (5th ed.). Springer.
- Garman, M.B., Klass, M.J. (1980). On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1), 67–78.
- Guyon, J., Lekeufack, J. (2023). Volatility is (mostly) path-dependent. *Quantitative Finance*, 23(9), 1221–1258.

References (2/3)

- ▣ Hamilton, J.D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384.
- ▣ Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.
- ▣ Inclán, C., Tiao, G.C. (1994). Use of cumulative sums of squares for retrospective detection of changes of variance. *Journal of the American Statistical Association*, 89(427), 913–923.
- ▣ Jarque, C.M., Bera, A.K. (1980). Efficient tests for normality, homoscedasticity and serial independence of regression residuals. *Economics Letters*, 6(3), 255–259.
- ▣ Jobson, J.D., Korkie, B.M. (1981). Performance hypothesis testing with the Sharpe and Treynor measures. *Journal of Finance*, 36(4), 889–908.
- ▣ Kim, T.-H., White, H. (2004). On more robust estimation of skewness and kurtosis. *Finance Research Letters*, 1(1), 56–73.
- ▣ Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- ▣ Ljung, G.M., Box, G.E.P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- ▣ Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- ▣ Lobato, I., Nankervis, J.C., Savin, N.E. (2001). Testing for autocorrelation using a modified Box–Pierce Q test. *International Economic Review*, 42(1), 187–205.
- ▣ Magdon-Ismael, M., Atiya, A.F., Pratap, A., Abu-Mostafa, Y.S. (2004). On the maximum drawdown of a Brownian motion. *Journal of Applied Probability*, 41(1), 147–161.

References (3/3)

- ▣ Memmel, C. (2003). Performance hypothesis testing with the Sharpe ratio. *Finance Letters*, 1(1), 21–23.
- ▣ Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- ▣ Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1), 61–65.
- ▣ Robinson, P.M. (1995). Gaussian semiparametric estimation of long range dependence. *Annals of Statistics*, 23(5), 1630–1661.
- ▣ Rogers, L.C.G., Satchell, S.E. (1991). Estimating variance from high, low and closing prices. *Annals of Applied Probability*, 1(4), 504–512.
- ▣ Sortino, F.A., van der Meer, R. (1991). Downside risk. *Journal of Portfolio Management*, 17(4), 27–31.
- ▣ Yang, D., Zhang, Q. (2000). Drift-independent volatility estimation based on high, low, open, and close prices. *Journal of Business*, 73(3), 477–492.